

Autonomous Forest Intelligence Drone

A PROJECT REPORT

SUBMITTED BY

SL NO.	NAME	UNIVERSITY ROLL NO.	STREAM
1.	ANIRBAN SAHA	430522010015	EIE 3 rd Year
2.	SOMSHUBHRA BOSE	430522010013	EIE 3 rd Year

Under the guidance of

Mrs. Sanghamitra Layek

Assistant Professor, Department of EIE

In

Electronics and Instrumentation Engineering

NARULA INSTITUTE OF TECHNOLOGY

81, Nilgunj Road, Agarpara, Kolkata, 700109

JUNE 2025

CERTIFICATE OF APPROVAL

This is to certify that the students mentioned below have submitted this thesis report on “solar charged seed spraying bot using mesh networking protocol”

The report is hereby ready for evaluation.

SL.NO.	STUDENT NAME	UNIVERSITY ROLL NO.	STREAM
01	Anirban Saha	430522010015	EIE – 3 RD Year
02	Somshubhra Bose	430522010013	EIE – 3 rd Year

Mrs. Sanghamitra Layek

(Supervisor)

Mrs. Sanghamitra Layek

(TEACHER-IN CHARGE)

DATE:

(EXTERNAL EXAMINER)

ACKNOWLEDGMENT

We would like to express our gratitude to Mrs. Sanghamitra Layek, Assistant Professor, Electronics and Instrumentation Engineering, Narula Institute of Technology, for her valuable guidance, which helped us a lot in due course of the project work.

We would like to thank, Mrs. Sanghamitra Layek, Teacher-in-Charge, Electronics and Instrumentation Engineering, Narula Institute of Technology, who had given all the support for performing experiments in this regard and extended all the facilities of the department for the project work.

<u>GROUP MEMBER NAME</u>	<u>UNIVERSITY ROLL NO.</u>	<u>SIGNATURE</u>
Anirban Saha	430522010015	
Somshubhra Bose	430522010013	

Abstract

Deforestation, illicit poaching, and forest fires are becoming more common, which has increased the demand for sophisticated and self-sufficient environmental monitoring systems. The Autonomous Forest Intelligence (AFI) Drone is an AI-powered unmanned aerial vehicle that was developed for ecological data collecting, anomaly identification, and real-time forest observation. Sensors such as the BMP280, DHT11, MPU6050, GPS, and MQ8 are integrated inside the drone to track motion, temperature, humidity, altitude, and hydrogen gas levels while in flight. Using RF modules, an ESP32 microcontroller gathers and sends sensor data, and an APM 2.8 flight controller permits autonomous waypoint missions and accurate navigation.

The system uses three deep learning models based on YOLOv8 to identify poachers, animals, and fire in real time. GPS-tagged detection results and captioned video feeds are shown on an intuitive dashboard via a web server driven by Flask. Functionalities that improve operating safety and use include real-time audio alarms, onboard SD card logging, and speech-activated parachute deployment. Additional assistance for the drone's autonomous operation comes from mathematical modeling of its kinematics, control systems, and battery consumption.

Environmental researchers and forest departments may now gather high-accuracy data and respond quickly with little ecological disruption thanks to this project's low-cost, modular, and scalable smart forest monitoring technology.

Keywords: AI surveillance, real-time detection, UAV, web dashboard, ESP32, environmental monitoring, YOLOv8, autonomous drone, forest fire detection, APM 2.8

Table of Contents

Section	Page No.
Certificate of Approval	i
Acknowledgment	ii
Abstract	iii
List of Figures	iv
Chapter 01 – Introduction	9
1.1 Background and Motivation	10
1.2 Problem Statement	11
1.3 Objectives of the Project	12
1.4 Scope of Work	13
1.5 Challenges in Forest Surveillance	14
Chapter 02 – Literature Review	15
2.1 Review of Drone-Based Surveillance	16
2.2 Related Research Projects	17-18
2.3 UAVs in Environmental Applications	19
2.4 AI in Forest Fire Detection	20
Chapter 03 – System Design and Architecture	21
3.1 Block Diagram of the Drone	22
3.2 Structural Design of the Drone	23
3.3 Circuit Diagram of the Drone	23
Chapter 04 – Component Details (Hardware)	24
4.1 APM 2.8 Flight Controller	25
4.2 NEO M8N GPS Module	26
4.3 BMP280 Sensor	27
4.4 DHT11 Sensor	28
4.5 FS-i6 Receiver	29
4.6 ESP32 DevKit V1	30
4.7 ESP32-CAM Module	31
4.8 MQ-8 Sensor	32
4.9 2200mAh Li-Po Battery	33
4.10 1000KV BLDC Motors	35
4.11 1045 Propellers	36
Chapter 05 – Software Integration	37
5.1 Website Architecture	38-39
5.2 Mission Planner	40
5.3 YOLO Object Detection Integration	41-42
5.4 Detection System Overview	43-44
5.5 Drone Camera Integration	45
5.6 Real-Time Alerts and Audio Feedback	46
5.7 GPS Integration for Location Logging	47
Chapter 06 – Datasets	48
6.1 Fire Detection Dataset	48
6.2 Poacher Detection Dataset	49

6.3 Animal Detection Dataset	50-51
Chapter 07 – Data Analysis and Results	52
7.1 Temperature & Humidity Graphs (DHT11)	53–60
7.2 MQ-8 Sensor Gas Readings	61-62
7.3 Battery Consumption Analysis	63-64
7.4 Fire, Animal, Poacher Detection Results	65–67
Chapter 08 - Applications and Use Cases	68
8.1 forest detection	69
8.2 illegal logging monitoring	69
8.3 Wildlife tracking	69
8.4 disaster area surveillance	70
8.5 smart forest management system	70
Chapter 09 - conclusion and future scope	71
References	72-73

List of figures

FIGURE NUMBER	DESCRIPTION	PAGE NO.
FIG 3.1.1	Block Diagram of the drone	21
FIG 3.2.1	Structural design of the drone	22
FIG 3.3.1	Circuit diagram of the drone	22
FIG 4.1.1	APM flight controller	24
FIG 4.1.2	NEO M8N Gps module	25
FIG 4.1.3	BMP280 Sensor	26
FIG 4.1.4	DHT 11 temperature and humidity sensor	27
FIG 4.1.5	fs-i6 receiver	28
FIG 4.1.6	ESP 32 devkit 1	29
FIG 4.1.7	ESP32 CAM module	30
FIG 4.1.8	MQ-8 Sensor	31
FIG 4.1.9	2200mmah battery	32
FIG 4.1.10	BLDC Motor	34
FIG 4.1.11	1045 propellers	35
FIG 5.1.1	Snapshot of the website designed for drone	37
FIG 5.2.1	Mission Planner Simulation Software	39
FIG 6.1.1	Examples of Augmented Images of fire dataset	48
FIG 6.2.1	Examples of Augmented Images of poacher dataset	49
FIG 6.3.1	Examples of Augmented Images of animal dataset	50
FIG 7.1.1.1	graph showing temperature vs days during the morning	53
FIG 7.1.1.2	graph showing temperature vs days in the afternoon	54
FIG 7.1.1.3	graph showing temperature vs days at night	55
FIG 7.1.2.1	graph showing humidity vs days in the morning	57
FIG 7.1.2.2	graph showing humidity vs days in the afternoon	58
FIG 7.1.2.3	graph showing humidity vs days at night	59
FIG 7.2.1	graph showing readings of the gas sensors at different parts of the day	61
FIG 7.3.1	graph showing the battery voltage vs time and power consumption vs time	63
FIG 7.4.1.1	(a) the image of fire before detection (b) the image of fire after detection	64
FIG 7.4.2.1	the training accuracy and loss curves of the trained fire model	64

FIG 7.4.2.1	(a) the image of animal before detection (b) the animal images after detection	65
FIG 7.4.2.2	the training accuracy and loss curves of the trained fire model	65
FIG 7.4.3.1	(a) the image of poacher before detection (b) the image of poacher after detection	66
FIG 7.4.3.2	the training accuracy and loss curves of the trained poacher model	66

CHAPTER 01

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Biodiversity, which refers to the variety of life across ecosystems, species, and genetic resources, is crucial for preserving the stability and health of our world. It promotes ecosystem services like as pollination, nutrient cycling, climate management, and food production. However, in recent decades, biodiversity has been diminishing at an alarming rate owing to human activity. The chief drivers of this loss are habitat degradation, deforestation, climate change, pollution, invasive species, wildfires, and rising human-animal conflicts. As ecosystems degrade, the extinction of species not only harms animals but also directly effects human well-being and sustainable development.

Conservation efforts have historically depended on conventional biodiversity monitoring methods such as manual field surveys, camera traps, and satellite photography. While effective, these strategies suffer significant drawbacks. Field surveys are time-consuming, labor-intensive, and frequently confined to accessible locations. Camera traps have limited geographical coverage and can overlook elusive or nocturnal animals. Satellite data, although vast in coverage, lacks the resolution and real-time responsiveness necessary for speedy decision-making and immediate danger identification. These issues underscore the urgent need for novel, efficient, and scalable monitoring solutions that may overcome these limits.

The Biodiversity Drone Project was started as a solution to these critical environmental issues. The idea is to deploy contemporary drone technology paired with artificial intelligence to boost biodiversity monitoring capabilities. By deploying drones equipped with high-resolution cameras, GPS modules, and AI-based species recognition algorithms, the initiative offers rapid, accurate, and non-intrusive ecological evaluations. This allows conservationists and academics to obtain real-time data from remote or vulnerable places without disrupting the natural ecosystem.

The initiative is not merely a technological innovation—it marks a step forward in sustainable environmental management. It intends to equip ecologists, forest departments, NGOs, and environmental academics with dependable tools for data-driven conservation. The decrease in labor effort, expense, and ecological harm makes this option extremely significant, especially in locations that are understudied or hard to access.

In summary, this research solves the constraints of existing monitoring techniques and corresponds with the global demand for smart, sustainable, and responsive conservation technology. It embodies the integration of electronics, environmental science, and artificial intelligence to preserve biodiversity and promote a resilient ecological future.

1.2 PROBLEM STATEMENT

Conserving biodiversity is a worldwide concern, although the methods used to monitor ecosystems remain mainly outmoded. Traditional techniques—such as field observations, camera traps, and satellite imaging—are frequently sluggish, resource-intensive, and restricted in geographical or temporal resolution. Manual surveys involve substantial human labor and expense and are tough to undertake in distant or dangerous regions. Camera traps, while non-intrusive, give only fragmentary information, sometimes failing to capture elusive or nocturnal animals. Meanwhile, satellite data lacks the precision needed for thorough, localized biodiversity research and cannot offer real-time danger detection.

These constraints result in data gaps that prevent timely conservation activities. In crucial cases like forest fires, poaching, or invasive species outbreaks, delayed reaction owing to poor monitoring tools can lead to irreparable ecological harm. Moreover, human presence during manual surveys may disrupt delicate environments, affecting wildlife behavior and lowering the accuracy of data.

To solve these difficulties, a smarter, quicker, and more ecologically responsible monitoring solution is urgently needed. The Biodiversity Drone Project was developed to bridge this gap—using drone-based surveillance driven by artificial intelligence to automate, speed, and increase the quality of biodiversity surveys while reducing environmental interference.

1.3 OBJECTIVES OF THE PROJECT

The Biodiversity Drone Project intends to build a comprehensive solution for biodiversity monitoring and ecological data collecting. The key objectives include:

- ✓ Design an AI-powered drone system capable of independently exploring varied situations.
- ✓ Capture high-resolution photos and videos for real-time wildlife and habitat monitoring.
- ✓ Integrate machine learning methods (e.g., YOLOv8) for accurate detection of animals, human activities, and environmental risks.
- ✓ Minimize ecological disruption by eliminating the requirement for on-site human presence.
- ✓ Provide decision-makers with actionable data for forest management, conservation planning, and policy formulation.
- ✓ Reduce operating expenses associated with manual surveys and long-term biodiversity research.
- ✓ Enable early identification of environmental risks including wildfires, deforestation, and habitat fragmentation.
- ✓ These aims correspond with worldwide conservation goals and illustrate the project's dedication to sustainability, innovation, and ecological integrity.

1.4 SCOPE OF WORK

The Biodiversity Drone Project includes hardware and software integration, field testing, and real-time analysis. The main areas of work are:

Drone Hardware Development: Building a quadcopter using the APM 2.8 flight controller, 1000KV brushless motors, GPS module, camera, and payload drop mechanism.

AI Integration: Using YOLOv8 object identification with OpenCV and TensorFlow to analyze pictures and find biodiversity-related traits.

Flight Control System: Mission Planner software is used to execute autonomous missions and optimize routes.

Data Acquisition and Storage: Obtaining and storing high-resolution photos and video material for processing and preservation.

Real-world deployment involves conducting field testing to assess detection accuracy, flight stability, and environmental effect.

Result Analysis and Feedback: Measuring performance parameters including battery life, detection rates, and deployment precision to improve the system.

The scope also involves investigating additional possible applications such as fire detection, agricultural monitoring, and urban green cover analysis.

1.5 CHALLENGES IN FOREST SURVEILLANCE

There are particular difficulties in monitoring biodiversity in wooded environments:

Dense Vegetation: Thick undergrowth and tree canopies obscure drone sensors and GPS signals, making it harder to see.

Remote and Inaccessible Terrain: Manual surveys are risky and ineffective in many ecologically rich locations because they are difficult to reach.

Weather Variability: Drone flying stability and camera performance may be hampered by rain, fog, and strong gusts.

animals Sensitivity: Mechanical and human disruptions can cause stress in animals, changing their usual habits and affecting the accuracy of data.

Limited Battery Life: In extensive forest settings, mission time and coverage area are limited by short flight durations.

Data Overload: Effective onboard and cloud-based computing solutions are needed to process massive amounts of real-time video and pictures.

Robust AI algorithms, lightweight designs, power-efficient systems, and intelligent navigation are necessary to meet these problems. The initiative includes ways to lessen these limitations without compromising the integrity of the environment.

CHAPTER 02

LITERATURE REVIEW

2.1 REVIEW OF DRONE BASED SURVEILLANCE

The usage of drones, also known as Unmanned Aerial Vehicles (UAVs), for surveillance has experienced a considerable surge over the past decade spanning varied industries such as defences, agriculture, disaster management, wildlife monitoring, and infrastructure inspection. Drone-based surveillance has changed traditional monitoring approaches by providing airborne data collecting in real-time, spanning enormous regions with great accuracy and decreased operating hazards. Their capacity to explore remote, hazardous, or challenging terrains makes them suitable for continuous monitoring in locations where human access is either risky or logistically infeasible.

The combination of cameras (visual, infrared, and thermal), sensors, GPS modules, and AI-based analytics has allowed drones to function as intelligent, autonomous systems for real-time observation and decision-making. These technologies are gradually replacing conventional surveillance techniques like static CCTV cameras, human patrols, and even satellite imagery because to their flexibility, mobility, and cost-effectiveness. In domains such as border surveillance, traffic monitoring, and search-and-rescue operations, drones give timely alerts and essential situational awareness that may save lives and resources.

In animal conservation and biodiversity monitoring, drone-based surveillance has played a significant role in minimizing human intervention, poaching, and habitat damage. Equipped with high-resolution and infrared imagery, drones can follow animal movement, identify nesting locations, and detect unlawful actions such as deforestation or encroachment. Surveillance drones are also widely employed for real-time fire detection and monitoring during natural catastrophes like forest fires, floods, or landslides, enabling authorities prepare quick reaction plans.

However, constraints such as limited flight time, data processing delays, privacy concerns, and the requirement for regulatory compliance continue to restrict wider deployment. Moreover, surveillance drones in sensitive regions must be properly calibrated to avoid ecological disruption.

Recent improvements focus on strengthening autonomy, incorporating deep learning models for object and pattern recognition, and improving drone swarm intelligence for cooperative surveillance missions. The future trajectory of drone-based surveillance leans towards more intelligent, energy-efficient, and ecologically conscientious systems.

As drone technology becomes more available and inexpensive, its applications in real-time monitoring, environmental protection, and security are likely to rise dramatically, opening the way for sustainable and non-intrusive surveillance methods.

2.2 RELATED RESEARCH PROJECTS

Numerous research initiatives worldwide have harnessed drone technology to promote environmental monitoring, animal conservation, and disaster management. These programs integrate UAVs with artificial intelligence, thermal imaging, and multispectral sensors to monitor ecological trends and give real-time insights for conservationists and academics.

One famous example is the Air Shepherd Program in Africa, which deploys drones outfitted with infrared cameras and GPS to detect poachers at night in national parks. This approach drastically decreased illicit hunting in places such as Zimbabwe and South Africa by providing aircraft observation without disrupting wildlife habitats. The drones utilized in this research operate independently and provide data to ground crews, enabling prompt action [1].

Another initiative, the Rainforest Connection (RFCx), employs drones and AI-powered acoustic sensors to identify illicit logging and deforestation in tropical forests. Although primarily focused on solar-powered audio sensors, the project developed to add aerial data to increase coverage. The real-time aspect of drone monitoring proved crucial in locations where human patrols were constrained by rugged terrain or enormous distances [2].

In India, the Forest Survey of India (FSI) began a drone-based pilot project to monitor forest degradation, tree cover loss, and afforestation activities. The study indicated that drones might supply data more often and with higher resolution than satellite photography. This program deployed UAVs with multispectral imagery and LiDAR to map forest density and analyze the success of conservation measures [3].

Additionally, academic research at organizations like ETH Zurich and MIT has proved the utility of swarming drones for environmental data collecting. These systems employ decentralized algorithms, allowing drones to monitor enormous regions collectively without overlapping, guaranteeing comprehensive and energy-efficient surveillance [4].

While these initiatives have highlighted the transformational potential of UAVs in environmental domains, they generally face restrictions such as restricted battery life, poor weather resistance, and limited onboard computing capabilities. However, continual developments in AI integration, solar-powered drones, and lightweight sensor technologies are tackling these limits.

By evaluating these studies, it becomes obvious that drone technology, when paired with sophisticated analytics and precision sensing, has enormous promise for conservation efforts. Lessons obtained from these pioneering experiments can immediately feed the development of more sophisticated, autonomous systems capable of proactive and large-scale ecological monitoring.

Several pioneering research studies have built the framework for UAV-based ecological and

environmental monitoring, constituting the backbone of contemporary drone-driven conservation methods. One notable initiative is the SMART (Spatial Monitoring and Reporting Tool) established in conjunction with WWF and the Wildlife Conservation Society. Though first conceived as a ground-based tool, its integration with drone technology has enabled geographically accurate surveillance of illicit activities, poaching concerns, and habitat degradation, allowing conservationists an automated strategy to safeguard biodiversity hotspots.

In the realm of forest fire detection, DETECTOR, an EU-funded project, blends drones with multi-spectral photography and AI algorithms for early forest fire detection in Mediterranean ecosystems. The experiment revealed that drone monitoring might detect fire breakouts faster than traditional tower or satellite-based systems, with superior geographical resolution and lower false alarms. Another unique initiative, DroneSEED, employs swarms of drones to rebuild forest ecosystems by spreading seed pods in fire-affected zones. This method not only speeds reforestation efforts but also illustrates how drone-based interventions may be both preventative and rehabilitative in environmental applications. In academic research, recent attempts by Zhang et al. (2021) deployed YOLOv4 in UAVs to locate endangered species in wildlife sanctuaries. Their findings demonstrated that deep learning algorithms may achieve excellent detection accuracy (>90%) for numerous animal groups utilizing thermal and visible spectrum imaging, paving the way for AI-enhanced ecological evaluations. These research initiatives, albeit different in scale, share a common goal—leveraging drone and AI technology to tackle significant environmental concerns. They serve as a precedent and validation for the Biodiversity Drone project, which intends to unite numerous functionalities—wildlife monitoring, fire detection, and ecological surveying—into a single, autonomous airborne device. While earlier initiatives frequently focus in a specific domain, the suggested method strives to provide a multipurpose tool to serve larger environmental goals efficiently.

2.3 UAVS IN ENVIRONMENTAL APPLICATIONS

Unmanned Aerial Vehicles (UAVs), or drones, have emerged as potent instruments in environmental science, giving unprecedented opportunities for remote sensing, ecological data gathering, and landscape analysis. Their mobility, cost-effectiveness, and ability to fly at low altitudes offer them a major edge over satellite imaging and piloted aircraft, especially in obtaining high-resolution pictures and real-time environmental data.

In the field of environmental and habitat monitoring, UAVs are used extensively to map forests, wetlands, and coastal zones. Through photogrammetry and LiDAR sensors, researchers may construct 3D landscape models to study changes in land cover, plant health, and habitat fragmentation across time. For instance, Koh and Wich (2012) demonstrated the use of drones in Indonesian rainforests to monitor orangutan numbers and record deforestation patterns, indicating how UAVs might assist biodiversity protection through non-intrusive surveillance [1].

In agriculture, UAVs are used to monitor crop health, irrigation levels, and insect infestations using multispectral and thermal photography. This sort of precision agriculture helps farmers to optimize input consumption, ultimately enhancing production and lowering environmental stress. A research by Zhang and Kovacs (2012) demonstrated how UAVs equipped with multispectral sensors may identify nitrogen deficits and detect early indicators of crop disease [2].

UAVs also serve a crucial part in disaster response and climate research. During wildfires, floods, or storms, drones give real-time views to emergency responders, assisting in efficient resource deployment and evacuation planning. In glacial zones, UAVs have been deployed to track ice melt and sea-level rise indicators. For example, researchers from the British Antarctic Survey employed UAVs to monitor penguin populations and glacier retreat, enabling safer and more frequent data collecting in difficult settings [3].

In air and water quality evaluation, UAVs mounted with chemical sensors can detect pollution hotspots, quantify particle matter, and evaluate water contamination. This use is particularly beneficial in industrial zones, oil spills, or post-disaster circumstances where traditional monitoring poses threats to human operators.

As regulatory frameworks improve, the incorporation of UAVs into long-term environmental policy is becoming more viable. Future trends include growing usage of drone swarms, AI-powered analytics, and solar-powered UAVs for continuous environmental observation. These developments strive to produce sustainable, intelligent, and automated systems capable of protecting ecosystems and minimizing environmental damage.

2.4 AI IN FOREST FIRE DETECTION

Forest fires pose a tremendous danger to ecosystems, biodiversity, and human populations. Traditional fire detection systems, such as watchtowers and satellite monitoring, generally suffer from delayed detection, poor resolution, and dependency on human attention. In recent years, Artificial Intelligence (AI) has emerged as a disruptive technology, enabling automatic and precise detection of forest fires at early stages. When coupled with UAVs, thermal imaging, and environmental sensors, AI systems greatly boost situational awareness and reaction capabilities.

AI-powered fire detection systems typically rely on image recognition models, notably Convolutional Neural Networks (CNNs), trained on datasets of smoke, flame patterns, and heat signatures. These algorithms analyze pictures or video frames collected from drones, CCTV cameras, or satellites to identify possible fire breakouts in real-time. The YOLO (You Only Look Once) algorithm, a state-of-the-art object identification framework, is commonly utilized in drone systems for fire spotting because to its rapid and efficient multi-class detection capabilities. A research by Muhammad et al. (2020) indicated that a YOLO-based drone system achieved over 90% detection accuracy in real-world forest situations [1].

In addition to object identification, AI models are employed for environmental forecast and risk assessment. Machine learning techniques such as Random Forest, SVMs, and LSTM (Long Short-Term Memory) networks examine historical data—temperature, humidity, wind speed, vegetation density—to anticipate fire-prone zones. These prediction techniques aid in pre-positioning firefighting resources and undertaking risk-informed forest management.

Thermal infrared (TIR) sensors placed on UAVs, when paired with AI analytics, provide a strong tool for real-time thermal anomaly identification, even under dense canopy cover or during darkness. This feature is particularly valuable for locating concealed ignition hotspots or smoldering flames that could go undiscovered by the human eye or visible-spectrum cameras.

The incorporation of AI also eliminates human mistake and permits continuous, automated surveillance across wide terrains. Moreover, AI systems may be implemented in multi-drone situations to coordinate detection and tracking activities efficiently. Despite its potential, the efficacy of AI in fire detection depends greatly on the quality and diversity of training datasets, as well as real-time processing capability. Therefore, hybrid techniques integrating edge computing, cloud analytics, and UAV-based data collecting are gaining appeal for scalable implementation.

CHAPTER 03

SYSTEM DESIGN AND ARCHITECTURE

3.1 BLOCK DIAGRAM

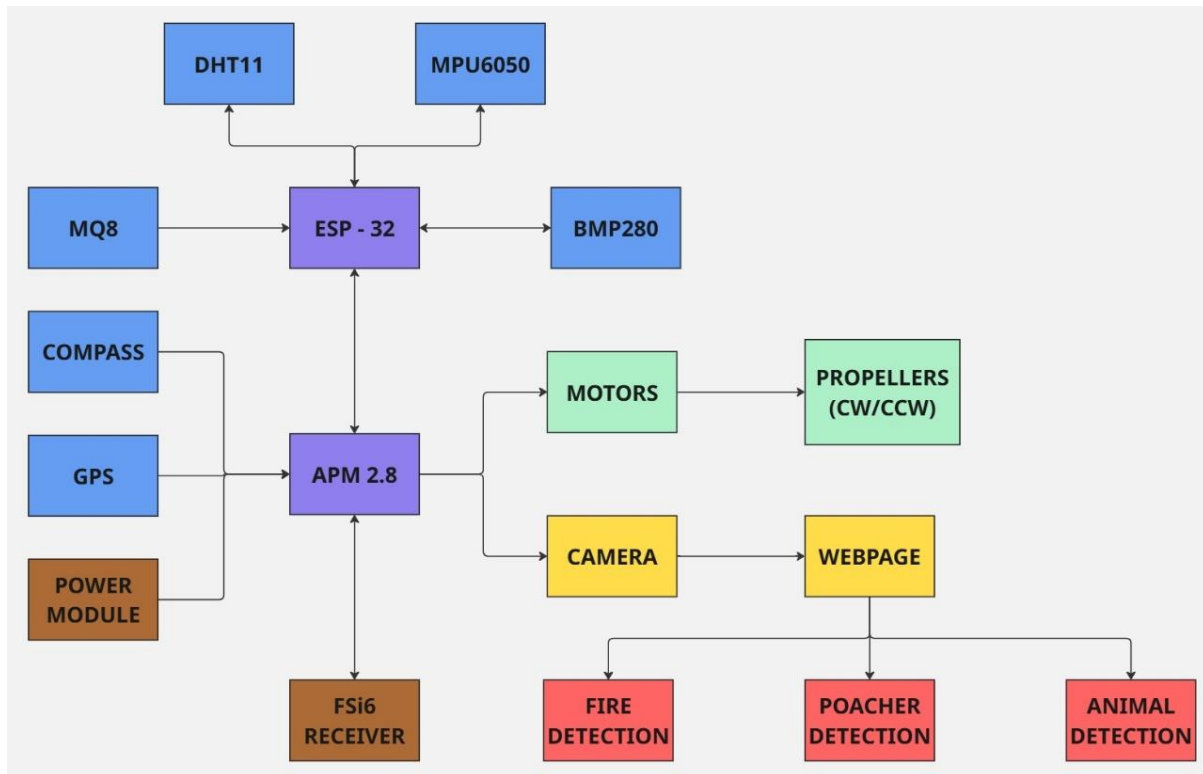


Fig 3.1.1 Block Diagram of the drone

A clever autonomous drone system that combines sensors, control modules, and artificial intelligence is depicted in this block diagram. Environmental sensors such as the DHT11 (temperature/humidity), BMP280 (altitude/pressure), MPU6050 (motion), and MQ8 (gas detection) provide data to the ESP32 microcontroller. Both environmental monitoring and navigation are supported by these readings.

Drone stability is controlled by the APM 2.8 flight controller, which also collects GPS and compass data for navigation and regulates the CW/CCW motors and propellers for flying. The FS-i6 receiver enables manual remote control, and the power module supplies and monitors power. For data exchange and real-time control, the ESP32 and APM interact.

Live video is recorded by a camera and sent to a web-based user interface (GUI), which further shows sensor data. The drone can react intelligently in forest settings thanks to AI modules for fire, poaching, and animal recognition that are hosted on this interface. This system provides a comprehensive solution for animal conservation, monitoring, and autonomous surveillance.

3.2 DRONE STRUCTURAL DESIGN

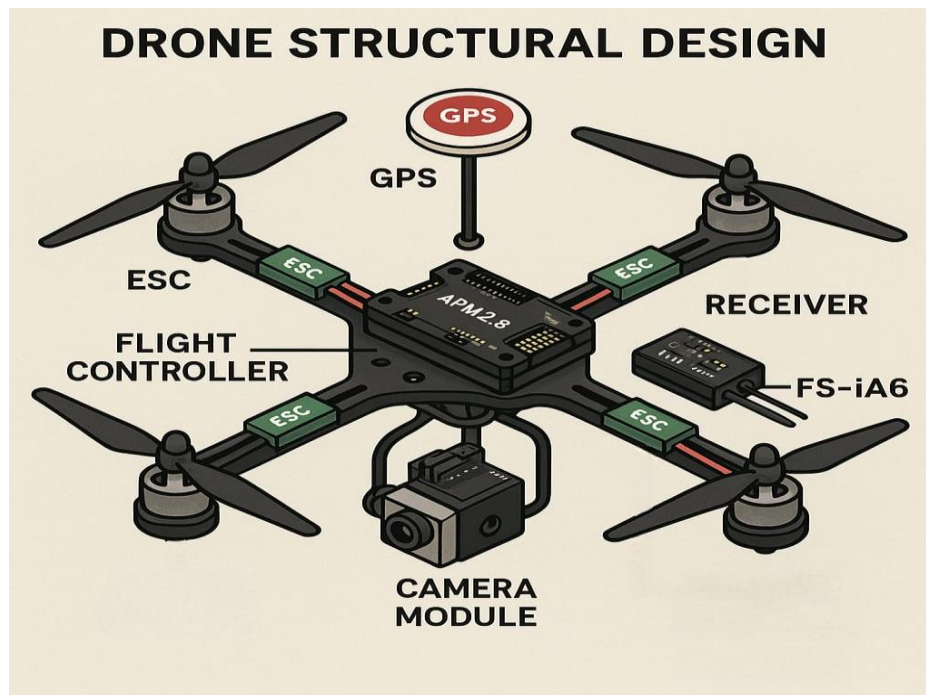


Fig 3.2.1 Structural design of the drone

3.3 CIRCUIT DIAGRAM

Fig 3.3.1 Circuit diagram of the drone

CHAPTER 04
COMPONENTS DETAILS
(HARDWARE)

4.1 COMPONENT SELECTION AND SPECIFICATIONS

1. APM 2.8 FLIGHT CONTROLLER

The APM 2.8 Flight Controller is the primary processing unit of my drone, responsible for stabilizing flight, interpreting sensor data, conducting autonomous missions, and guaranteeing reliable connection with the base station. In my project, which incorporates the Autonomous Forest Intelligence (AFI) drone, the APM 2.8 works as the brain of the system, providing precision navigation and real-time response to environmental data. The controller enables multiple flying modes including manual, stabilized, altitude hold, and fully autonomous GPS-guided flights. With support for ArduPilot firmware, the APM 2.8 enables significant customizability, making it excellent for mission-specific activities such as forest fire detection, GPS tracking, and aerial mapping.

Key Specifications of APM 2.8 Used in My Drone:

- ✓ Microcontroller: Atmel ATmega2560 @ 16 MHz
- ✓ IMU Sensors: 3-axis gyroscope, 3-axis accelerometer, 3-axis magnetometer
- ✓ Barometer: MS5611 high-resolution altimeter
- ✓ External GPS module support: Yes (attached via UART)
- ✓ Power input: 5V via power module (with current and voltage sensors)
- ✓ Connectivity: 6 PWM outputs (for ESCs), telemetry port (attached to HC-12), I2C, UART



Fig 4.1.1 APM flight controller

Usage in My Drone: In my AFI drone, the APM 2.8 is placed on the center disk of a modular frame. It collects orientation data from the onboard MPU6050, height information from the BMP280, and location data from the GPS module. The HC-12 RF module delivers telemetry to a ground station. It also logs flight data to the onboard microSD card for post-flight study. Through Mission Planner, I create autonomous flight courses, establish geofences, and watch live sensor data. This lets the drone to execute activities like as following pre-defined pathways in wooded regions, obtaining sensor readings, and returning to home autonomously, all governed by the APM 2.8.

2. NEO M8N GPS MODULE :

The Ublox Neo-M8N GPS module serves a key role in my Autonomous Forest Intelligence (AFI) drone by delivering precise location, altitude, and velocity data required for autonomous navigation and telemetry tracking. This GPS module, when connected with the APM 2.8 flight controller, allows the drone to conduct GPS-guided missions, return-to-home capabilities, and waypoint-based path following with precision.

Unlike prior GPS modules, the M8N enables multi-GNSS reception, which means it can simultaneously use GPS, GLONASS, Galileo, and QZSS satellite systems, substantially boosting signal lock speed and accuracy even in thick forest conditions or under partial canopy cover.

Key Specifications of M8N GPS Used in My Drone:

- ✓ GNSS Support: GPS, GLONASS, Galileo, QZSS
- ✓ Update Rate: Up to 10 Hz
- ✓ Horizontal Position Accuracy: ± 2.5 meters
- ✓ Cold Start Time: ~26 seconds
- ✓ Hot Start Time: ~1 second
- ✓ Power Supply: 3.3V–5V (compatible with APM)
- ✓ Communication Interface: UART (TX/RX), I2C also available
- ✓ Integrated Compass: HMC5883L (for orientation data)
- ✓ Antenna: Active ceramic patch antenna with ground plane
- ✓ Backup Battery: Yes, maintains satellite data for rapid locking



Fig 4.1.2 NEO M8N Gps module

Usage in My Drone: Mounted at the top of the drone's structure for optimal sky visibility, the M8N GPS module connects to the APM 2.8 via the specialized GPS connector. The onboard HMC5883L compass provides heading data, helping the drone to retain orientation during mission-critical operations like forest mapping or wildfire monitoring. The GPS data allows the drone to independently travel through waypoints, perform loiter operations, and execute RTL (Return to Launch) when contact is lost or the mission is done. Its precision enables safe and efficient area coverage during environmental scanning operations.

3. BMP280

The BMP280 is a high-precision barometric pressure and temperature sensor used in my Autonomous Forest Intelligence (AFI) drone to assess altitude and air pressure. It serves a key role in maintaining flight stability, especially in missions that need accurate altitude hold, terrain following, or atmospheric data collecting in forest situations. This sensor delivers real-time data to the APM 2.8 flight controller, allowing the drone to determine its height above sea level with better precision than GPS alone. Unlike ultrasonic sensors, the BMP280 is tiny, light-weight, and not impacted by plant or ground reflectivity—making it perfect for airborne applications.

Key Specifications of BMP280 Used in My Drone:

- ✓ Pressure Range: 300 to 1100 hPa (hectopascal)
- ✓ Altitude Accuracy: ± 1 meter
- ✓ Temperature Range: -40°C to $+85^{\circ}\text{C}$
- ✓ Operating Voltage: 1.8V to 3.6V (uses logic-level shifter for 5V compatibility)
- ✓ Interface: I2C (used in my drone), also supports SPI
- ✓ Power Consumption: Ultra-low (excellent for battery-powered devices)
- ✓ Data Output Rate: Up to 157 Hz
- ✓ Size: $\sim 3.6\text{mm} \times 3.8\text{mm}$ (extremely tiny)

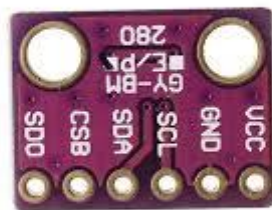


Fig 4.1.3 BMP280 Sensor

Usage in My Drone: The BMP280 is placed on the sensor disk of my drone and linked to the APM 2.8 via the I2C connection. It continually delivers barometric data that the flight controller utilizes to estimate vertical speed, regulate altitude, and perform altitude hold and auto takeoff/landing tasks.

In forest missions, this sensor helps the drone fly safely across uneven terrain by altering its height in real-time based on pressure measurements. It also captures atmospheric data (pressure and temperature) for environmental research and fire risk analysis. Together with GPS altitude, the BMP280 boosts total vertical accuracy. Unlike GPS, which can vary by several meters, the BMP280 delivers greater resolution, necessary for low-altitude forest flights.

4. DHT11

The DHT11 is a digital temperature and humidity sensor placed into my Autonomous Forest Intelligence (AFI) drone to monitor environmental variables during flight. This sensor is vital

for gathering real-time weather data, which is particularly significant for forest health monitoring, fire risk prediction, and microclimate study in distant places. Although it is not as precise as industrial-grade sensors, the DHT11 is lightweight, low-power, and easy to connect with microcontrollers, making it perfect for incorporation into tiny aerial platforms like my drone.

Key Specifications of DHT11 Used in My Drone:

- ✓ Temperature Range: 0°C to 50°C
- ✓ Temperature Accuracy: $\pm 2^{\circ}\text{C}$
- ✓ Humidity Range: 20% to 90% RH (Relative Humidity)
- ✓ Humidity Accuracy: $\pm 5\%$ RH
- ✓ Sampling Rate: 1 reading per second (1 Hz)
- ✓ Interface: Single-wire digital interface
- ✓ Power Supply: 3.3V to 5.5V (5V in my system)
- ✓ Current Consumption: ~ 2.5 mA during data transmission
- ✓ Size: Compact (about. 15.5mm x 12mm)

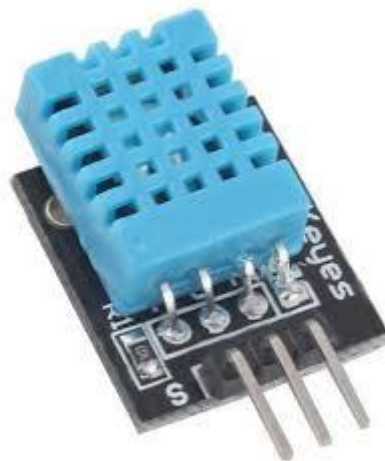


Fig 4.1.4 DHT 11 temperature and humidity sensor

Usage in My Drone:

The DHT11 is installed on the sensor disk of the drone in an exposed location to guarantee accurate environmental readings. It is coupled to the ESP32 microcontroller, which reads the temperature and humidity measurements and logs them into the microSD card or broadcasts them via the HC-12 RF module for real-time ground station monitoring. This environmental data is timestamped and geotagged using GPS data, giving a rich dataset for climate trend analysis, forest ecosystem health checks, and fire-prone zone detection. Although rudimentary, the DHT11's data complements other sensors like the BMP280 and GPS module, offering a more full image of the forest environment.

5. FSI6 RECEIVER

The FlySky FS-i6 Receiver (FS-iA6 or FS-iA6B) is a 6-channel 2.4GHz radio receiver used in my Autonomous Forest Intelligence (AFI) drone to create manual RC (Remote Control) connection between the drone and the operator. It operates as the communication bridge for manual override, testing, emergency control, and pre-mission calibration.

While my drone is built for autonomous forest missions, the FS-i6 receiver gives a dependable fallback alternative. In case of mission failure, GPS loss, or terrain complexity, I may switch to manual mode and safely navigate or land the drone using my FlySky FS-i6 transmitter.

Key Specifications of FS-i6 Receiver Used in My Drone:

- ✓ Frequency: 2.4 GHz (AFHDS 2A protocol)
- ✓ Channels: 6 (supports all fundamental drone controls - throttle, pitch, roll, yaw, mode switch, etc.)
- ✓ Voltage Range: 4.0V – 8.4V (powered by APM 2.8)
- ✓ Failsafe Function: Yes (auto RTL or motor cut in signal failure)
- ✓ Antenna: Dual antenna diversity for steady signal reception
- ✓ Binding: Easy connecting with FlySky FS-i6 Transmitter
- ✓ Output Interface: PWM (linked to input channels on APM 2.8)



Fig 4.1.5 fs-i6 receiver

Usage in My Drone:

The FS-i6 receiver is attached to the input channels of the APM 2.8 flight controller. Each channel is assigned to distinct controls:

- Channel 1 – Roll
- Channel 2 – Pitch
- Channel 3 – Throttle
- Channel 4 – Yaw
- Channel 5 – Flight mode changeover
- Channel 6 – Auxiliary functions (e.g., parachute release or camera)

During preflight tests, I utilize the FS-i6 transmitter to manually test motor responses, calibrate ESCs, and verify control surfaces. In-flight, it lets me to select between Stabilize, Loiter, and Auto modes or recover control if the autonomous mission hits difficulties or failure.

6. ESP32

The ESP32 is a powerful dual-core microcontroller that includes Wi-Fi and Bluetooth. It acts as the secondary processing unit in my Autonomous Forest Intelligence (AFI) drone. While the APM 2.8 handles flight stability and autonomous navigation, the ESP32 is in charge of data collecting, processing, communication, and environmental parameter tracking. The ESP32's multitasking capabilities enable it to take data from sensors like as the BMP280, DHT11, and MPU6050, analyze it, and either log it to a microSD card or communicate it to the ground station via the HC-12 RF module.

Key Features of the ESP32 Used in My Drone:

- ✓ Processor: Dual-core Tensilica LX6 at up to 240 MHz.
- ✓ RAM: 520 KB SRAM.
- ✓ Flash memory: 4MB (expandable).
- ✓ Wireless: Integrated 2.4 GHz Wi-Fi and Bluetooth (BLE).
- ✓ GPIOs: More than 30 programmable pins.
- ✓ Interfaces include I2C, SPI, UART, ADC, and PWM.
- ✓ Operating voltage: 3.3 volts.
- ✓ Power consumption: Low-power modes are provided.
- ✓ Microcontroller Board: ESP32 DevKit V1 (used for my drone)



Fig 4.1.6 ESP 32 devkit 1

The ESP32 serves as a data manager and communicator in the drone. Here's how.

- ✓ *Sensor Integration:* The DHT11 measures temperature and humidity, the BMP280 measures pressure and altitude, and the MPU6050 measures motion via I2C.
- ✓ *Data logging:* Saves sensor data with timestamps and GPS coordinates (from the GPS module) to an onboard microSD card module.
- ✓ *Wireless Transmission:* The HC-12 RF module transmits real-time sensor data and telemetry to a ground-based monitoring system.
- ✓ *Parachute Control:* Receives verbal orders or emergency signals and activates a servo-based parachute deployment mechanism.
- ✓ *Failsafe Systems:* Can monitor power levels and environmental risks and send directives to APM in critical situations.

7. ESP 32 CAM

The ESP32-CAM module is a tiny, low-cost microcontroller with a built-in camera, and in my Autonomous Forest Intelligence (AFI) drone, it acts as the visual sensor unit. Its major duty is to record real-time aerial photos and broadcast live video, which is vital for forest surveillance, fire detection, animal monitoring, and post-mission analysis. With its compact size and integrated Wi-Fi, the ESP32-CAM delivers sophisticated edge vision capabilities to the drone without adding much weight or power consumption, making it perfect for lightweight aerial systems.

Key Specifications of ESP32-CAM Used in My Drone:

- ✓ Processor: Dual-core Tensilica LX6 @ 240 MHz
- ✓ Camera: OV2640 (2MP)
- ✓ Resolution: up to 1600x1200 (UXGA)
- ✓ Format: JPEG, RGB565
- ✓ RAM: 520 KB SRAM + 4 MB PSRAM
- ✓ Storage: MicroSD card support up to 32GB
- ✓ Connectivity: 2.4 GHz Wi-Fi (802.11 b/g/n)
- ✓ GPIO: 10 useable pins for peripherals (e.g., sensors or servos)
- ✓ Power Input: 5V regulated
- ✓ Size: ~27mm x 40.5mm
- ✓ Antenna: Onboard or external (IPEX connection)



Fig 4.1.7 ESP32 CAM module

Usage in My Drone:

The ESP32-CAM is installed on the front or underbelly of the drone providing an excellent view of the environment. It fulfills numerous critical functions:

- a) Image Capture: Captures high-resolution photos at regular intervals throughout flight. These are kept on the microSD card together with GPS locations and timestamps for further analysis.
- b) Live Video Streaming: Sends low-latency video over Wi-Fi (or through a Wi-Fi repeater module on the ground) during short-range missions.
- c) Event Detection (optional with AI models): Can be designed to detect fire-like colors (red/orange blobs), smoke, or movement using lightweight onboard computer vision models, and trigger warnings.
- d) Environmental Logging: Syncs with the main ESP32 board to tag photos with sensor data (temperature, humidity, pressure).

In conjunction with the drone's GPS and flight controller, the ESP32-CAM turns the drone into a low-cost aerial surveillance tool capable of both human and automated visual monitoring. It considerably boosts the drone's situational awareness and post-mission analytical capacity.

8. MQ8 SENSOR :

The MQ-8 is a gas sensor module designed to detect hydrogen (H_2) gas in the environment. In my Autonomous Forest Intelligence (AFI) drone, the MQ-8 is used as part of an air quality monitoring system, especially to detect hydrogen gas leaks, combustion byproducts, or abnormal chemical levels that may indicate fire outbreaks, industrial pollution, or chemical plant leakage near forest areas. Though normally utilized in industrial safety applications, incorporating the MQ-8 onto a drone offers it the unique ability to fly over hazardous or inaccessible regions, gathering hydrogen concentration data in real time.

Key Specifications of MQ-8 Used in My Drone:

- ✓ Target Gas: Hydrogen (H_2)
- ✓ Detection Range: 100 – 10,000 ppm
- ✓ Output Signal: Analog voltage and digital threshold output
- ✓ Heater Voltage: $5V \pm 0.2V$ (for internal sensing element)
- ✓ Response Time: <10 seconds
- ✓ Preheat Time: 20 seconds (minimum); 24–48 hours for precise use
- ✓ Operating Temperature: $-10^{\circ}C$ to $50^{\circ}C$
- ✓ Power Consumption: Moderate (because to integrated heater)
- ✓ Interface: Analog output linked to ESP32 ADC pin



Fig 4.1.8 MQ-8 Sensor

Usage in My Drone: The MQ-8 sensor is installed on the sensor disk of the drone, with its sensing element exposed to ambient air. It is interfaced with the ESP32 microprocessor, which reads the analog voltage output from the sensor, indicating hydrogen concentration in the air.

Key functions in the drone include:

- a) Hydrogen Leak Detection: In forest regions near industrial or power plants, the drone may autonomously fly through possible gas leak zones and record hydrogen levels.

- b) Fire and Combustion Sensing: Detects early levels of hydrogen, which can be generated during certain combustion processes—an indirect signal of forest fires or chemical reactions.
- c) Data Logging and Telemetry: The ESP32 logs hydrogen levels together with GPS position, timestamp, and other sensor data to a microSD card and broadcasts real-time alarms via the HC-12 RF module.

9. BATTERY

My Autonomous Forest Intelligence (AFI) drone is powered by an 11.1V 2200mAh LiPo (Lithium-Polymer) battery, which acts as the primary power source for both flying and onboard electronics. This type of battery is frequently utilized in drone applications because to its high energy density, small weight, and ability to generate high discharge currents essential for motors and sensors. The 3-cell (3S) arrangement produces a nominal voltage of 11.1V (3.7V per cell), with a fully charged voltage of 12.6V and a minimum safe voltage of roughly 9.6V.

Key Specifications of the Battery:

- ✓ Battery Type: LiPo (Lithium Polymer)
- ✓ Voltage: 11.1V (3S)
- ✓ Capacity: 2200mAh (2.2 Ah)
- ✓ Energy: 24.42 Wh ($11.1V \times 2.2 \text{ Ah}$)
- ✓ Discharge Rate: 25C–35C (varies with brand)
- ✓ At 25C, peak discharge = $2.2A \times 25 = 55A$
- ✓ Weight: ~180–200 grams
- ✓ Connector: XT60 (used in my drone)
- ✓ Charge Time: ~1 hour with 2.2A balancing charger
- ✓ Cycle Life: ~200–300 charge cycles (with adequate maintenance)



Fig 4.1.9 2200mmah battery

Usage in My Drone:

This battery powers:

- a) The APM 2.8 flying controller
- b) Brushless motors with ESCs
- c) ESP32 and ESP32-CAM
- d) Sensors (BMP280, DHT11, MPU6050, MQ8)
- e) GPS Module with HC-12 RF transmitter
- f) Servo mechanism for parachute deployment
- g) The battery is coupled to a Power Module, which gives clean, regulated power to the APM and monitors current, voltage, and power usage, providing in-flight battery health monitoring.
- h) On average, the 2200mAh battery gives 8–12 minutes of flying duration, depending on:
Payload weight , Wind conditions, Motor power draw, Sensor activity and video streaming load ,For safety, the drone features a low-voltage failsafe, triggering a Return-to-Launch (RTL) or auto-land when the voltage lowers around 10.5V.

10. 1000KV MOTOR

In my Autonomous Forest Intelligence (AFI) drone, I employ 1000KV Brushless DC (BLDC) motors as the main propulsion components. These motors are crucial for ensuring consistent lift, mobility, and economical power consumption, especially in a quadcopter arrangement built for autonomous aerial observation in wooded environments. The “1000KV” rating implies the motor rotates at 1000 RPM (Revolutions Per Minute) per volt provided without any load. So at 11.1V (from my 3S LiPo battery), each motor can spin up to around 11,100 RPM, making it excellent for medium-lift multirotor drones like mine.

Key Specifications of 1000KV BLDC Motor Used in My Drone:

- ✓ KV Rating: 1000 KV
- ✓ Input Voltage: 7.4V – 14.8V (best at 11.1V)
- ✓ Recommended Propeller Size: 9x4.5 or 10x4.5 inches (used: 1045)
- ✓ Max Current: ~13–15 A (per motor)
- ✓ Thrust Output: ~800–900g per motor (with 1045 propellers at 11.1V)
- ✓ Weight: ~70–80g (each)
- ✓ Shaft Diameter: 3.17 mm
- ✓ Mounting Holes: Standard quadcopter frame compatibility
- ✓ Bearing Type: Dual ball bearing
- ✓ Construction: Outrunner motor (stator inside, magnets outside)



Fig 4.1.10 BLDC Motor

Usage in My Drone:

The drone employs four 1000KV BLDC motors in an X-configuration (quadcopter). Each motor is linked with:

- a) A 10x4.5 propeller for best thrust and efficiency
- b) A 30A ESC (Electronic Speed Controller) provides precise control
- c) A motor mount on the lightweight carbon/plastic frame
- d) The motors get PWM control signals from the APM 2.8 flight controller via the ESCs.

During flight:

- a) The motors vary their speed dynamically for pitch, roll, yaw, and throttle control
- b) The push created balances the drone in air and allows it to climb, fall, hover, and move in all directions

In conjunction with a 2200mAh 11.1V LiPo battery, the 1000KV motors give a reasonable mix between thrust and power efficiency, supporting up to 1–1.2 kg of total payload, including the flight controller, sensors, ESP32 modules, and camera system.

11. 1045 PROPELLER

The 1045 propeller (often referred to as “10x4.5”) is a vital component in the propulsion system of my Autonomous Forest Intelligence (AFI) drone. These propellers are combined with the 1000KV brushless DC motors to give the essential lift, stability, and thrust efficiency for flight in forest situations where hovering, slow-speed maneuvering, and load-bearing capabilities are crucial. The numerals "1045" represent the propeller's size: 10 inches in diameter 4.5 inches pitch (distance it would go forward in one full spin under ideal conditions)

Key Specifications of 1045 Propeller Used in My Drone:

- ✓ Diameter: 10 inches (254 mm)
- ✓ Pitch: 4.5 inches (114.3 mm)

- ✓ Material: ABS plastic (often used), occasionally reinforced with nylon or carbon fiber
- ✓ Weight: ~12–15 grams apiece
- ✓ Blade Count: 2
- ✓ Rotation: Available in clockwise (CW) and counter-clockwise (CCW) pairs
- ✓ Mounting: Compatible with 3.17mm or 4mm motor shafts (adapter rings provided)



Fig 4.1.11 1045 propellers

Usage in My Drone:

The 1045 propellers are installed on the four 1000KV motors in an X-configuration quadcopter frame, with two CW and two CCW propellers to balance torque and provide reliable yaw control.

CHAPTER 05

SOFTWARE INTEGRATION

5.1 Website Architecture :

The architecture of the website is developed as an integrated real-time wildlife monitoring system that blends powerful AI-based detection algorithms with a user-friendly online interface. The system is constructed on a multi-tier architecture consisting of a frontend web interface designed using HTML, CSS, and JavaScript, and a backend server written with the Flask Python framework. The backend runs three different YOLO (You Only Look Once) deep learning models for fire, animal, and poacher detection, processing live video feeds and presenting detection results dynamically to the frontend.

Frontend Layer

The frontend encompasses the essential components of web technology — HTML provides the structure and semantic layout of the webpage, including regions allocated for video display, notifications, and species count visualization. CSS designs the website, offering a logical and visually pleasant user experience with responsive design features for clear presentation of detection findings, such as bounding boxes and alarms. JavaScript enables the dynamic nature of the site, managing asynchronous data fetching, video stream rendering, and updating detection overlays in real-time without the need to reload the page.

Through JavaScript, the website accesses live video feeds supplied by the backend Flask server as MJPEG streams and overlays detection information right on the video channel. JavaScript also handles user interactions and visual indications for detected events like fire warnings, animal species count, and poacher alerts, offering a smooth and engaging monitoring experience.

Backend Layer:

At the core of the backend sits Flask, a lightweight Python web framework that manages HTTP requests, provides video streams, and incorporates the AI models for detection. The backend initializes and maintains video capture from an IP camera or default webcam, processing frames constantly through three YOLO models:

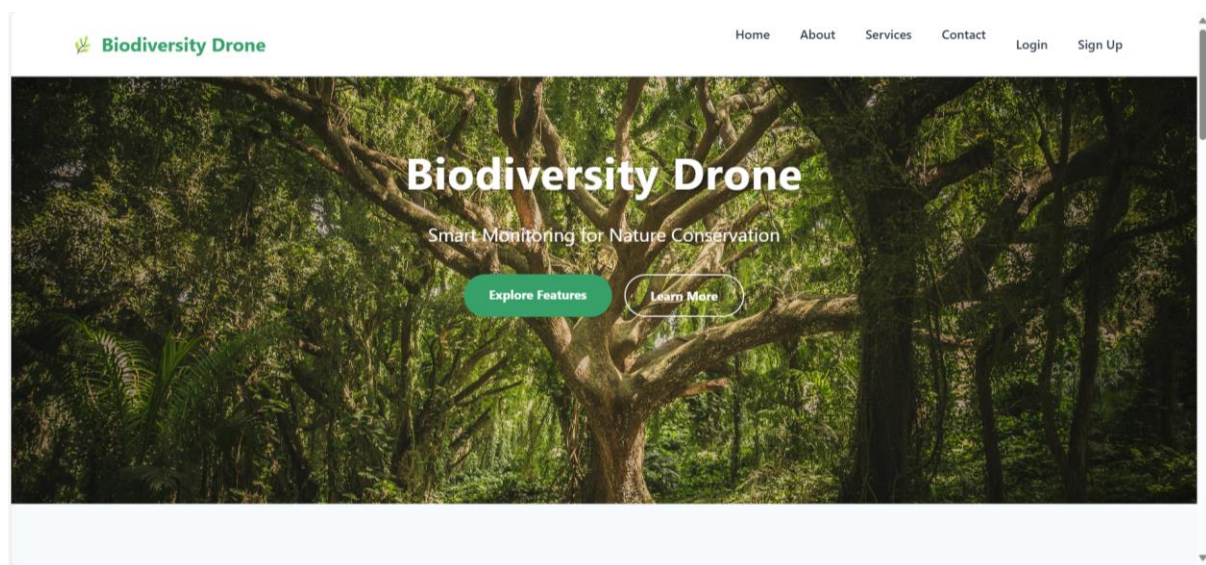


Fig 5.1.1 Snapshot of the website designed for drone

Fire Detection Model (best.pt):

This model examines each video frame to identify fire or smoke, employing confidence and Intersection over Union (IoU) thresholds to eliminate false positives. Upon detecting fire, the backend files the event with timestamp and geolocation data collected via IP-based API requests. It also collects and retains photos with annotated bounding boxes for subsequent examination or reporting.

Animal Detection Model (yolov8n.pt):

This lightweight YOLO model recognizes numerous animals from a given list of species relevant to the natural habitat. The model generates bounding boxes and class labels for observed animals. The backend accumulates counts of each species in real-time and overlays this information on the camera feed. This permits biodiversity monitoring and population tracking remotely.

Poacher Detection Model:

This component employs two YOLO models — one for person detection and another custom-trained model (weapon.pt) for weaponry. The technology examines video frames for human presence along with identification of weapons such as knives or firearms. If overlap between a person and a weapon bounding box is identified, the algorithm signals a probable poacher warning. This detection enables proactive security actions to preserve wildlife.

Data Handling and Streaming Flask feeds the processed video frames with detection overlays as continuous MJPEG streams, which the frontend consumes and displays. This streaming approach offers near real-time display of ongoing detections without page reloads. The backend controls caching, frame buffering, and error handling to enable stable video transmission even in changing network circumstances.

Logs created from fire detection events, including timestamps and GPS positions, are stored to CSV files for historical study. Snapshot photos are kept with unambiguous bounding box annotations, which may be utilized for documentation and subsequent AI model training or validation.

Integration and User Experience

Together, the frontend and backend comprise a tightly integrated system where the user may monitor environmental conditions, animal activity, and security concerns smoothly using a web browser. The architecture enables modularity, enabling upgrades or replacements of detection models without rebuilding the frontend interface. This flexibility means that the platform can adapt with improvements in AI and camera technology.

.

5.2 Mission Planner :

Mission Planner is a widely used open-source ground control software built primarily for configuring and maintaining drones, especially quadcopters. It acts as a complete interface between the user and the drone's flight controller, enabling setup, calibration, monitoring, and mission planning with simplicity.

One of the key tasks of Mission Planner is its ability to calibrate numerous sensors and components crucial for the stable and precise flying of a quadcopter. These include accelerometers, gyroscopes, magnetometers (compass), radio transmitters, and ESCs (Electronic Speed Controllers). Proper calibration ensures that the drone reads sensor data accurately, maintains balance, and responds precisely to control inputs, which is vital for safe and reliable operation.

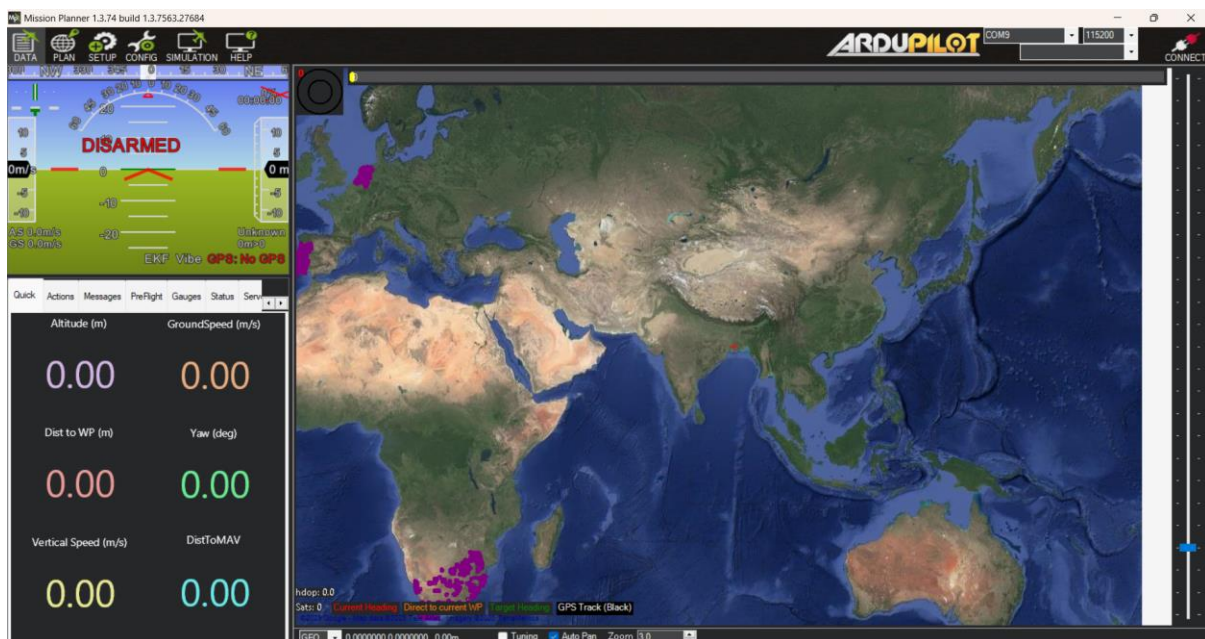


Fig 5.2.1 Mission Planner Simulation Software

Mission Planner provides a user-friendly graphical interface that leads customers step-by-step through each calibration procedure. For example, during accelerometer calibration, the user is prompted to put the drone in several orientations so the program may capture baseline sensor data. Similarly, compass calibration entails turning the drone in various directions to correct magnetic interference, guaranteeing precise heading information.

Besides calibration, Mission Planner includes additional capabilities such as real-time flight monitoring, waypoint mission planning, parameter adjustment, and firmware upgrades. It provides communication with the drone by telemetry radios or USB connections, allowing users to plan autonomous flights by setting GPS waypoints, altitude, speed, and other flying characteristics.

Overall, Mission Planner is a crucial tool for drone fans and experts alike, simplifying the complicated duties of sensor calibration and flight management, hence boosting the performance, dependability, and safety of quadcopter drones.

5.3 Integration of YOLO Object Detection Model :

5.3.1 Introduction

You Only Look Once (YOLO) is a set of real-time object identification systems based on convolutional neural networks. First proposed by Joseph Redmon et al. in 2015, YOLO has experienced multiple revisions and enhancements, becoming one of the most popular object identification frameworks.

The term "You Only Look Once" alludes to the fact that the method takes only one forward propagation run through the neural network to create predictions, unlike earlier region proposal-based algorithms like R-CNN that require thousands for a single image.

5.3.2 YOLO Architecture Overview

YOLO splits the input picture into a $S \times S$ grid. Each grid cell is responsible for predicting:

- ✓ B bounding boxes: (x, y, w, h) (x,y,w,h)
- ✓ Objectness score
- ✓ Class probabilities

This architecture allows YOLO to execute detection and classification concurrently, enabling faster inference speed critical for drone-based real-time applications.

5.3.3 Mathematical Formulation of YOLO

1. Bounding Box Prediction

Each bounding box is defined as:

$$(x, y, w, h)$$

Where:

x, y : coordinates of the center of the box (relative to the grid cell)

w, h : width and height of the box (normalized by the picture size)

Each forecast box also carries a confidence score:

$$\text{Confidence} = P(\text{object}) \cdot \text{IoU}_{\text{pred, truth}}$$

where :

$P(\text{object})$: probability of object presence

$\text{IoU}_{\text{pred, truth}}$: Intersection-over-Union with ground truth box

2. Class Score Calculation

The final class score for each recognized object is computed as:

$$\text{Class Score} = P(\text{object}) \cdot P(\text{class } i | \text{object})$$

Where $P(\text{class } i | \text{object})$ is the conditional class probability.

5.3.4 Sample Output (Detection Confidence Table)

Table 1 Output of the yolo model

Frame No.	Object Detected	Confidence	Coordinates (x, y, w, h)
17	Smoke	0.91	(0.45, 0.37, 0.18, 0.22)
18	Person	0.87	(0.67, 0.59, 0.20, 0.35)

5.4 Detection System Overview

The biodiversity monitoring drone system uses the capabilities of YOLOv8 (You Only Look Once, version 8)—a cutting-edge real-time object identification framework. YOLOv8 is noted for its excellent accuracy, speed, and efficiency, making it perfect for deployment on resource-constrained devices like drones. The model architecture blends convolutional neural networks (CNNs) with anchor-free detection algorithms to boost performance while minimizing latency. Three separate YOLOv8 models are employed in the system, each trained for a specialized task:

Fire Detection Model: The fire detection system in our research utilizes computer vision techniques employing a deep learning model termed best.pt, which has been particularly trained on a comprehensive fire dataset. This dataset comprises a wide range of fire scenarios — from small flames and early ignition stages to large-scale wildfires and smoke present — confirming the model’s ability to identify fire in diverse environmental conditions and illumination circumstances.

The model runs on each video frame acquired by the drone’s onboard or IP camera in real time. It evaluates visual clues such as flame color, texture, and movement patterns to effectively categorize fire occurrences. Once fire is identified, the algorithm annotates the frame with a bounding box, label, and confidence score.

This early fire detection capacity is critical in forest areas because timely intervention can minimize the development of wildfires and decrease ecological harm. By combining this detection process with real-time alarms, GPS tracking, and snapshot saving, the system guarantees that all fire-related occurrences are both swiftly recognized and documented.

Overall, this model represents a significant aspect of the drone’s environmental monitoring capabilities, delivering an automated, rapid, and intelligent approach for wildfire identification and mitigation utilizing advanced computer vision techniques.

Animal Detection Model: The animal identification component of our biodiversity drone system is powered by the YOLOv8n model, a lightweight yet highly efficient variant of the YOLO (You Only Look Once) family tailored for real-time applications. This model has been fine-tuned to recognize a specific collection of 10 animal classes: ['bird', 'cat', 'dog', 'horse', 'sheep', 'cow', 'elephant', 'bear', 'zebra', 'giraffe']. These species encompass a broad spectrum of wildlife, including both domestic and wild animals, with a concentration on large mammals and endangered species typically observed in wooded or protected zones.

Each shot acquired by the drone’s camera is processed through this model to determine the presence of any of the target species. The detections are labeled with bounding boxes, class names, and confidence ratings. Using collections. Counter, the system maintains a running count of creatures observed with each flight, enabling biodiversity monitoring and population

tracking over time.

This capability is especially beneficial for forest rangers, conservationists, and researchers, since it provides automated observation of animal movement patterns, possible threats to species, and the influence of environmental changes. The model promotes ecological understanding through real-time and historical data, facilitating proactive animal preservation actions.

Poacher Detection Model: The poacher detection system is constructed by merging two independent YOLO models to identify possible hazards in the wild. The first model is the lightweight YOLOv8n (yolov8n.pt), which is used particularly for identifying human presence in the drone's camera feed. The second model (weapon.pt) is a custom-trained YOLO model based on a handpicked weapon dataset. This variant is specialized in identifying deadly things such as knives, firearms, or other portable weapons.

Rather of depending on a single monolithic model, our dual-model technique enhances both modularity and accuracy. Each frame from the drone's video stream is initially run through the person identification model to identify persons. The same frame is then evaluated by the weapon detection model to see if any recognized weapons are present.

By linking the physical closeness of observed humans and weapons inside a frame, the system may estimate whether an individual is likely carrying a weapon, hence identifying them as a prospective poacher. This strategy decreases false positives by requiring the co-occurrence of both person and weapon detections, which is more reliable than training a single model on restricted armed-person data.

This system provides real-time and precise identification of poachers, letting forest officials respond rapidly to illegal hunting risks and increasing overall wildlife protection and surveillance.

5.5 Drone Camera Integration

The key input to the biodiversity drone monitoring system is a live video feed acquired from the drone's onboard camera. This camera might be a high-resolution FPV module or an external IP camera put on the drone body. The solution is designed to provide flexibility in camera hardware to ensure adaptation across different drone platforms.

The video stream is incorporated into the software using the OpenCV package, notably the `cv2.VideoCapture()` method. This function offers real-time frame-by-frame access to the live video stream. If an IP camera is utilized, the stream URL (typically RTSP or HTTP) is provided into VideoCapture. For onboard cameras (e.g., CSI cameras on Raspberry Pi or USB webcams), numerical indices (such 0 or 1) are used to connect directly.

Each frame recorded from the stream works as an input to the object detection process. The system processes frames in real time, allowing it to respond promptly to any identified events such as fire, animal sightings, or possible poachers.

In addition to continuous video processing, the system also includes a paused snapshot mode, which allows individual frames to be taken and stored at any point for comprehensive analysis or manual review. These snapshots are kept locally with timestamped filenames, making it easier to monitor when and where certain events happened.

This dual functionality—real-time processing and snapshot capture—makes the device robust, adaptable, and extremely effective for both surveillance and data logging in wildlife monitoring missions.

5.6 Real-Time Alerts and Audio Feedback

To enable fast reaction during wildlife monitoring missions, the biodiversity drone system contains both text-to-speech (TTS) and audio alarm methods for real-time input. These features increase situational awareness, especially when visual monitoring of the video feed is not feasible.

5.6.1 Pyttsx3 (Text-to-Speech Engine)

The system leverages the pyttsx3 Python package, a sophisticated offline text-to-speech engine that translates alarm messages into spoken English. Whenever a detection is made—such as a fire, animal, or weapon—the system generates a matching notification (e.g., “Fire detected!”, “Animal detected!”, or “Poacher detected!”). These messages are then vocalized using pyttsx3. This audio output guarantees that the operator or team is quickly notified even when they are not looking at the display, allowing them to reply rapidly. Since pyttsx3 functions without internet reliance, it is appropriate for field operations in distant places.

5.6.2 Pygame Audio Alerts

In addition to speech input, the system adds different sound cues utilizing the pygame mixer module. Specific sound files such as alert.wav or beep.mp3 are activated when different sorts of items are recognized. Each detecting type has a distinct tone:

Fire Detection: Triggers an urgent, high-pitched beep to alert danger.

Animal Detection: Plays an informative tone, indicating observation-worthy events.

Weapon Detection: Emits an emergency alert tone, demanding urgent attention.

These layered acoustic systems provide redundant and reliable alerting, important for speedy decision-making in key biodiversity zones.

6.7 GPS Integration For Location Logging

Accurate location data is vital for monitoring and tracking animal activities and incidents such as fires or poacher presence. In this system, GPS coordinates are merged directly from the IP camera's inbuilt GPS module to offer real-time geographic information for each detection event. The IP camera continually broadcasts video together with its current latitude and longitude coordinates included in the metadata or available via its API. This GPS data is retrieved programmatically throughout the detection phase to correlate any recognized object—fire, animal, or weapon—with accurate position information.

Whenever a detection happens, the system stores the GPS coordinates together with a timestamp, providing a geotagged record of the incident. This permits later investigation and mapping of hotspots, animal migration patterns, or poacher activity zones. Since the IP camera offers direct GPS data, there is no need for external methods such as web scraping or using third-party IP location services, which are typically imperfect and dependent on internet access. Additionally, this strategy removes reliance on magnetic readings or human heading predictions, simplifying the system and enhancing dependability.

The logged GPS data may be utilized for many downstream applications, including showing detections on geographic information system (GIS) platforms, giving specific alarm locations to enforcement teams, or preserving full event records for conservation initiatives. This seamless incorporation of location data considerably boosts the system's practical value in real-world biodiversity monitoring.

CHAPTER 06

DATASETS

6.1 Fire Detection Dataset

To improve the performance of our fire detection model, we used 12 different picture augmentation techniques on the original dataset. These augmentations aid in simulating real-world variances and improving the model's generalizability to various fire conditions. Starting with simple geometric changes, we used horizontal flip, vertical flip, rotation (30° and 90°), and shift-scale to change the orientation and location of fire in the photos. These augmentations allow the model to become insensitive to the angle and location of firing.

We also employed blur and Gaussian noise to simulate picture deterioration caused by motion or poor camera quality, allowing the model to identify fire even at low clarity. Brightness, contrast, and hue-saturation modifications were made to accommodate for lighting circumstances and color fluctuations that are common in outdoor settings or at different times of the day. Furthermore, elastic transformation caused modest bending in the flame structure, making the model more resilient to dynamic and erratic fire forms. These modifications are critical since fire does not have a consistent form or pattern.

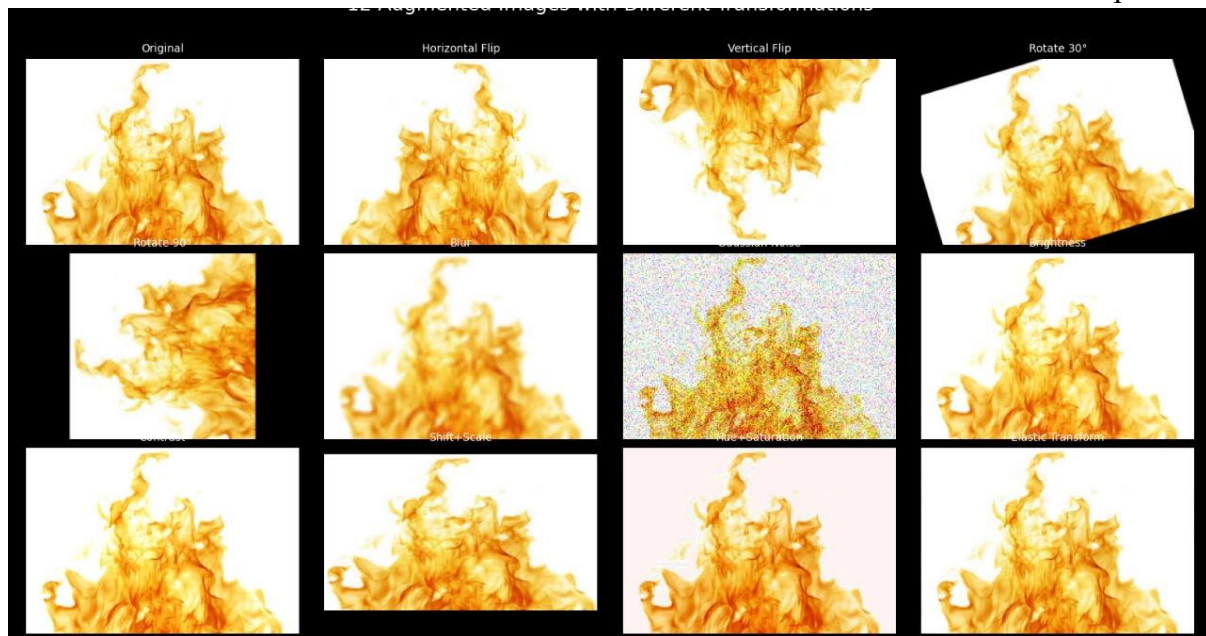


Fig 6.1.1 Examples of Augmented Images of fire dataset

Overall, this substantial augmentation method improves the model's capacity to detect fire reliably in a variety of real-world settings, enhancing safety and response systems in surveillance applications.

6.2 Poacher Detection Dataset

We employed 12 extensive picture augmentation approaches to improve the resilience of our animal detection algorithm in a variety of wildlife habitats. These augmentations reflect real-world differences in illumination, posture, and backdrop, ensuring that the model is well-suited

to a variety of field circumstances. To account for changes in animal orientation and location, we first used spatial augmentations such as horizontal and vertical flips, rotation (30° and 90°), and shift-scale. These changes aid the model in identifying animals regardless of how they look in the frame.

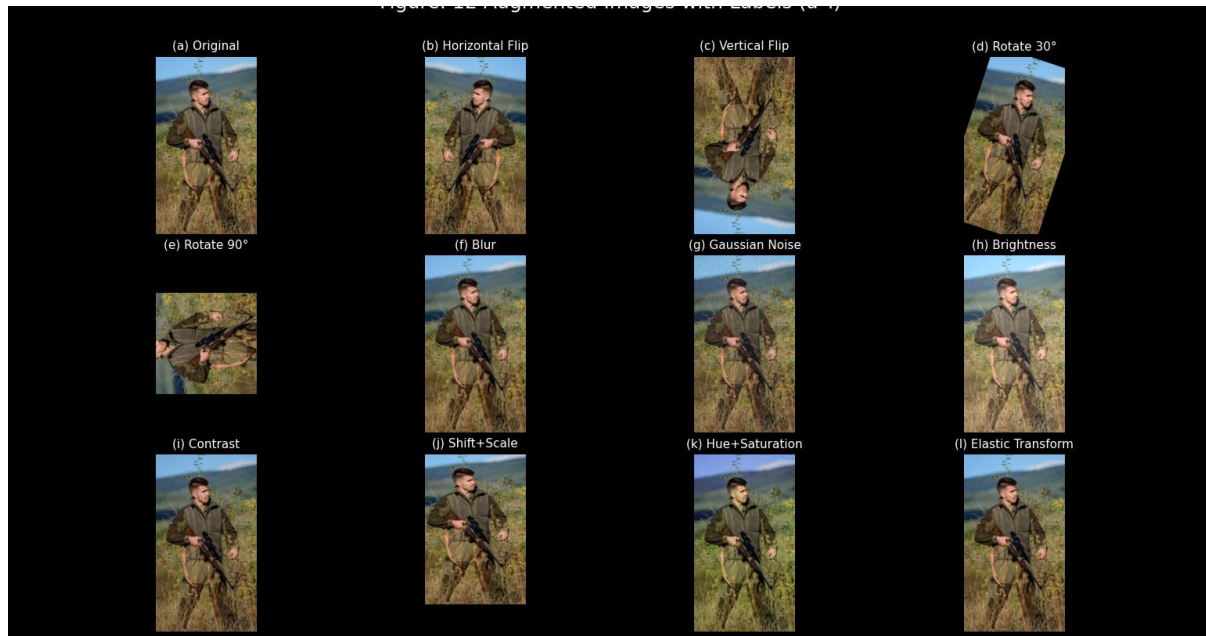


Fig 6.2.1 Examples of Augmented Images of poacher dataset

Next, we employed blur and Gaussian noise to replicate motion blur or picture distortion produced by low-quality cameras or environmental factors. This equips the model to recognize animals under less-than-ideal visual conditions. Brightness, contrast, and hue-saturation adjustments improved the model's ability to handle various lighting circumstances, such as day-night transitions or forest shadows. Finally, elastic transformation was employed to replicate minor deformations in animal forms caused by movement or partial vegetation occlusion. These enhancements ensure that the model can recognize creatures of varying sizes, postures, and appearances in complicated environments. This augmentation method enhances detection accuracy and reliability, making the system suitable for real-time biodiversity monitoring in dynamic animal environments.

6.3 Animal Detection Dataset

We used 12 data augmentation strategies to train a more flexible and accurate poacher detection model, which entails detecting persons and weapons in a variety of settings. Because poachers might appear in a variety of postures, lights, and circumstances, augmentation is critical to increasing resilience. Geometric transformations such as horizontal and vertical flips, rotation (30° and 90°), and shift-scale were employed to represent the many body postures and angles that a person or weapon may appear at in real video. This allows the model to detect subjects even when they are partially rotated or away from the center of the picture.

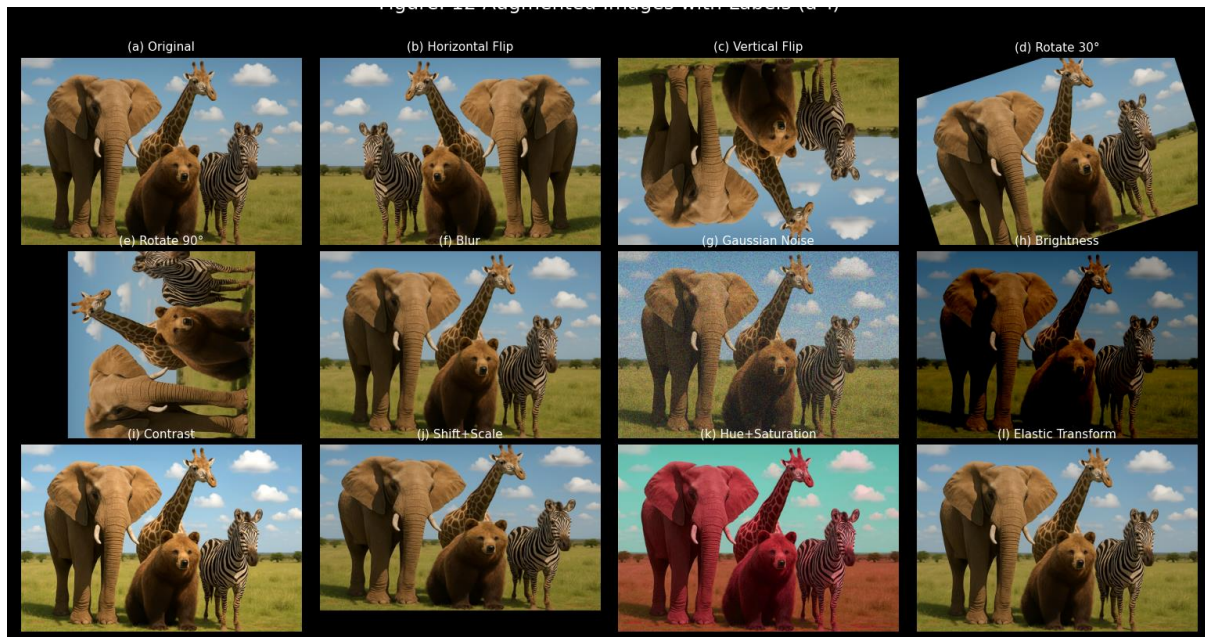


Fig 6.3.1 Examples of Augmented Images of animal dataset

To solve environmental inconsistencies, we used blur and Gaussian noise to replicate low-resolution video feeds, motion blur, and sensor noise. Furthermore, brightness, contrast, and hue-saturation adjustments were made to simulate changing weather, time of day, and camera sensors in real-world circumstances.

The elastic transformation was utilized to gently alter the contour of the human body or weapon outline, improving the model's capacity to identify hidden or partially apparent threats. These augmentations enhance the poacher detection model, leading to improved security and threat identification in forest monitoring systems.

CHAPTER 07
DATA ANALYSIS AND RESULTS

7.1 Environmental Sensor Data

7.1.1 Temperature Sensor (DHT-11) :

The AFI drone is equipped with a temperature and humidity sensor which can record environmental data. The data was recorded in three times a day i.e. morning, afternoon and night.

Morning (7am – 8am) :

Table 2 : atmospheric temperature data collected in the morning

Date	Temperature (°C)
11-May	28.4
12-May	27.5
13-May	27.4
14-May	28.3
15-May	27.9
16-May	28.4
17-May	28.3
18-May	28
19-May	27.5
20-May	27.1
21-May	27.7
22-May	28.5
23-May	28.1
24-May	27.3
25-May	28.1
26-May	28.2
27-May	27.5
28-May	28.1
29-May	28
30-May	27.3
31-May	28.2
01-Jun	28.3
02-Jun	28.6
03-Jun	28.6
04-Jun	27.2
05-Jun	27.4
06-Jun	28.3
07-Jun	28.3
08-Jun	28.3
09-Jun	30.3
10-Jun	28.3
11-Jun	28.7

The above table shows that the temperature data recorded during the morning time ranged between 27°C – 30°C.

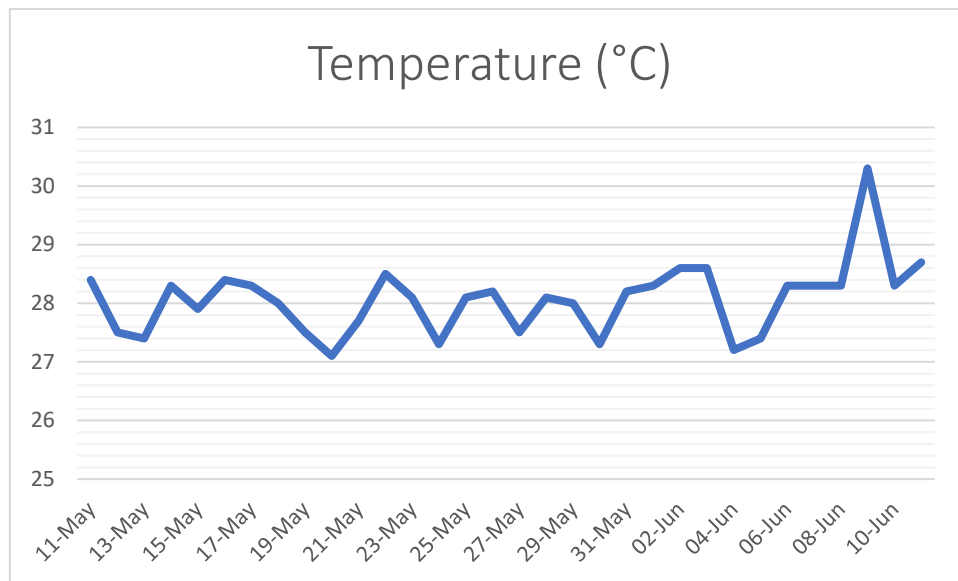


Fig 7.1.1.1 graph showing temperature vs days during the morning

Afternoon (1 pm – 3 pm) :

Table 3 : atmospheric temperature data collected in the afternoon

Date	Temperature (°C)
11-May	35.7
12-May	36.3
13-May	36.2
14-May	36
15-May	37.7
16-May	36.8
17-May	35.6
18-May	37.1
19-May	38
20-May	37.2
21-May	35.4
22-May	36.2
23-May	37.4
24-May	36
25-May	36.8
26-May	37
27-May	35.9
28-May	36.5
29-May	34.2
30-May	35.8

31-May	36.3
01-Jun	35.6
02-Jun	37.6
03-Jun	35.5
04-Jun	36.2
05-Jun	36.6
06-Jun	37.5
07-Jun	35.5
08-Jun	37.3
09-Jun	36.5
10-Jun	35.8
11-Jun	36.8

The above table shows that the temperature data recorded during the afternoon time ranged between 34°C – 38°C.

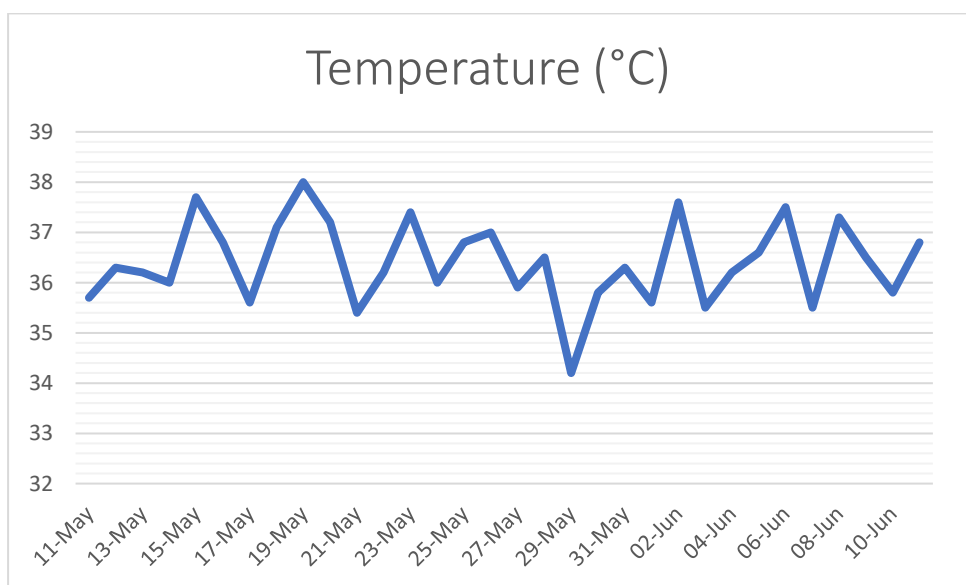


Fig 7.1.1.2 graph showing temperature vs days in the afternoon

Night (8pm – 10pm) :

Table 4 : atmospheric temperature data collected at night

Date	Temperature (°C)
11-May	29.3
12-May	28.9
13-May	28.9
14-May	29.7
15-May	29.5
16-May	29.5

17-May	29.8
18-May	29
19-May	29.4
20-May	28.8
21-May	29.2
22-May	28.9
23-May	29.1
24-May	29.4
25-May	28.5
26-May	30.3
27-May	28.4
28-May	28.3
29-May	29.7
30-May	29.5
31-May	29.4
01-Jun	29.4
02-Jun	29
03-Jun	28.5
04-Jun	29
05-Jun	28.6
06-Jun	29.6
07-Jun	28.9
08-Jun	28.5
09-Jun	28.8
10-Jun	29.2
11-Jun	28.7

The above table shows that the temperature data recorded during the afternoon time ranged between 28°C – 29°C.

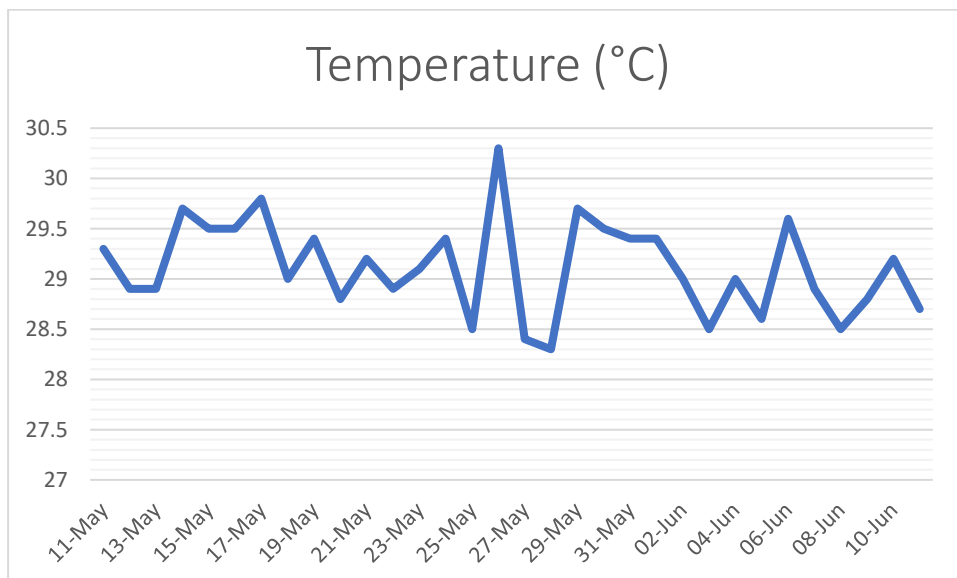


Fig 7.1.1.3 graph showing temperature vs days at night

7.1.2 Humidity Sensor :

The AFI drone is equipped with a temperature and humidity sensor which can record environmental data. The data was recorded in three times a day i.e. morning, afternoon and night.

Morning (7 am – 9am) :

Table 5 : humidity data collected in the morning

Date	Humidity (%)
11-May	87.9
12-May	87.3
13-May	85.4
14-May	87.5
15-May	84.5
16-May	85.5
17-May	85
18-May	86.2
19-May	90.6
20-May	82.3
21-May	87.4
22-May	82.8
23-May	85.1
24-May	88.2
25-May	86.1
26-May	83.8
27-May	84.6
28-May	87.4
29-May	84.5
30-May	86.4
31-May	86.1
01-Jun	84.7
02-Jun	90.3
03-Jun	87.3
04-Jun	81.9
05-Jun	86.4
06-Jun	84.7
07-Jun	87.7
08-Jun	84.4
09-Jun	85.8
10-Jun	87
11-Jun	87.7

The above table shows that the humidity data recorded during morning ranged between 81.9 – 90.

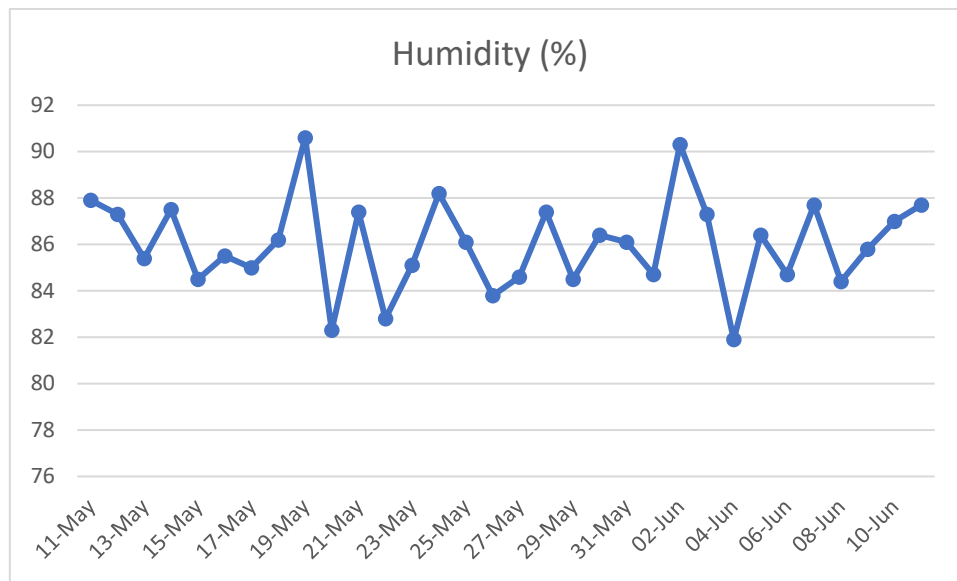


Fig 7.1.2.1 graph showing humidity vs days in the morning

Afternoon (1pm – 3 pm) :

Table 6 : humidity data collected in the afternoon

Date	Humidity (%)
11-May	53.5
12-May	51.5
13-May	53.2
14-May	52
15-May	53.3
16-May	54.7
17-May	57
18-May	49.9
19-May	58.3
20-May	48.1
21-May	52.6
22-May	54.5
23-May	53.7
24-May	51.4
25-May	52.5
26-May	51.8
27-May	51.5
28-May	55.1
29-May	53.9
30-May	51.3
31-May	55.2

01-Jun	53.8
02-Jun	55
03-Jun	54.6
04-Jun	50.9
05-Jun	51.6
06-Jun	54.9
07-Jun	54.5
08-Jun	52.9
09-Jun	53.3
10-Jun	56.2
11-Jun	51.5

The above table shows that the humidity data recorded during the afternoon ranged between 50.9 – 56.2.

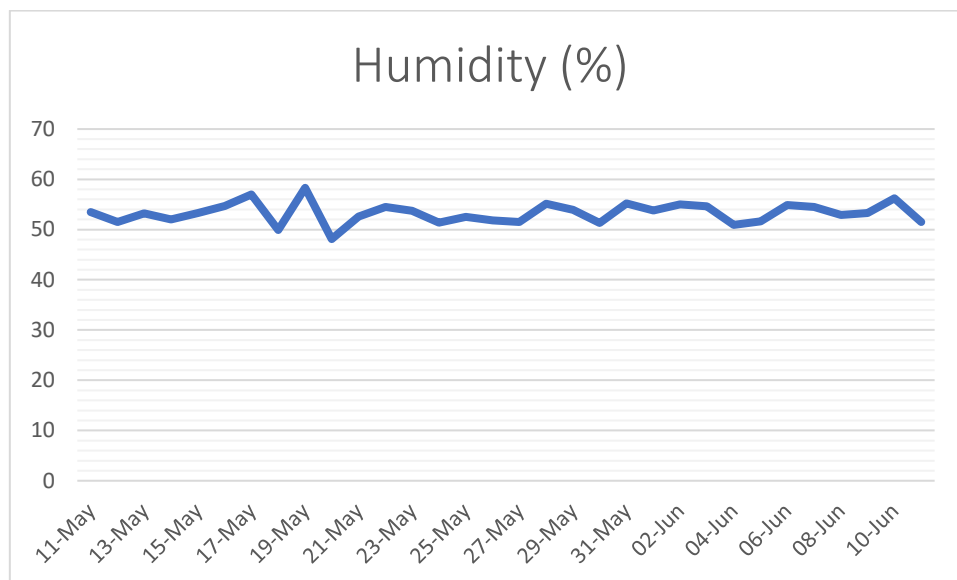


Fig 7.1.2.2 graph showing humidity vs days in the afternoon

Night (8pm – 10 pm) :

Table 7 : humidity data collected at night

Date	Humidity (%)
11-May	82.4
12-May	84.5
13-May	84.5
14-May	83
15-May	83.1
16-May	84.5
17-May	81.1
18-May	81.2
19-May	82.6

20-May	83.6
21-May	84.6
22-May	87
23-May	85.7
24-May	83.7
25-May	84
26-May	82
27-May	84
28-May	83.4
29-May	84.6
30-May	82.3
31-May	85
01-Jun	87.1
02-Jun	83.8
03-Jun	84.8
04-Jun	85.4
05-Jun	83.2
06-Jun	84.4
07-Jun	84
08-Jun	84.2
09-Jun	82.5
10-Jun	84
11-Jun	85

The above table shows that the humidity data recorded during the afternoon ranged between 81.9 – 90.

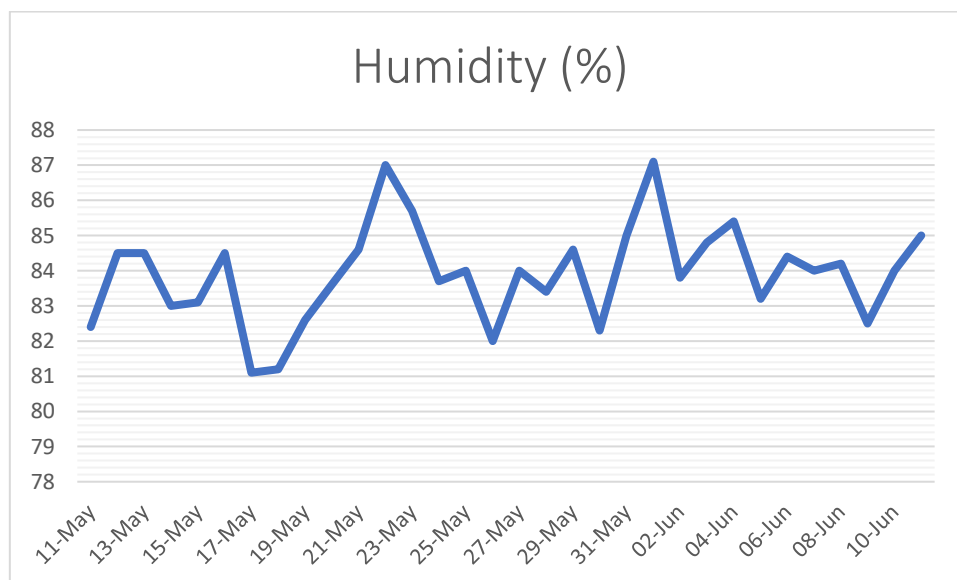


Fig 7.1.2.3 graph showing humidity vs days at night

7.2 MQ-8 Sensor data :

Table 8 : readings of the gas sensors at different parts of the day

Date	Morning (ppm)	Afternoon (ppm)	Evening (ppm)	Night (ppm)
11-May-25	3.1	5.4	4.8	3.5
12-May-25	3.3	5.2	4.7	3.6
13-May-25	3.0	5.5	5.0	3.8
14-May-25	3.5	6.1	5.2	4.0
15-May-25	3.2	5.8	5.1	3.9
16-May-25	3.4	5.6	4.9	3.7
17-May-25	3.1	5.3	4.6	3.4
18-May-25	3.0	5.1	4.5	3.3
19-May-25	3.2	5.4	4.8	3.6
20-May-25	3.3	5.6	5.0	3.8
21-May-25	3.1	5.2	4.7	3.5
22-May-25	3.5	5.9	5.3	4.1
23-May-25	3.6	6.0	5.4	4.0
24-May-25	3.4	5.8	5.1	3.9
25-May-25	3.2	5.5	5.0	3.8
26-May-25	3.1	5.2	4.6	3.5
27-May-25	3.0	5.0	4.4	3.3
28-May-25	3.3	5.7	4.9	3.7
29-May-25	3.4	5.9	5.2	3.9
30-May-25	3.5	6.1	5.4	4.1
31-May-25	3.3	5.8	5.0	3.8

01-Jun-25	3.2	5.6	4.9	3.7
02-Jun-25	3.1	5.3	4.7	3.5
03-Jun-25	3.4	5.7	5.0	3.9
04-Jun-25	3.2	5.4	4.8	3.6
05-Jun-25	3.0	5.1	4.5	3.3
06-Jun-25	3.1	5.2	4.6	3.4
07-Jun-25	3.3	5.6	4.9	3.8
08-Jun-25	3.4	5.9	5.2	3.9
09-Jun-25	3.5	6.2	5.5	4.1
10-Jun-25	3.3	5.8	5.1	3.8
11-Jun-25	3.2	5.5	4.9	3.6

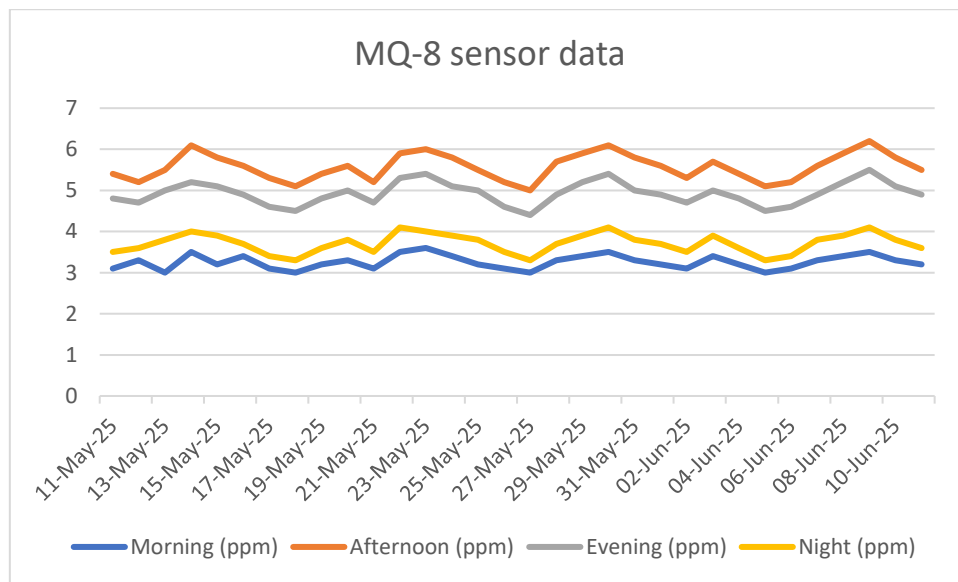


Fig 7.2.1 graph showing readings of the gas sensors at different parts of the day

7.3 Battery Consumption vs Flight Time

Table 9 : tabular comparison of the battery mah with battery voltage and time

Time (min)	Battery Voltage (V)	Current (A)	Power Used (mAh)	Remaining Capacity (mAh)
0	12.6	0.0	0	2200
1	12.3	5.2	86.7	2113.3
2	12.1	5.4	180.7	2019.3
3	11.9	5.3	269.2	1930.8
4	11.8	5.6	362.6	1837.4
5	11.7	5.7	457.3	1742.7
6	11.6	5.5	550.0	1650.0
7	11.5	5.4	640.0	1560.0
8	11.4	5.3	728.0	1472.0
9	11.3	5.2	813.3	1386.7
10	11.2	5.2	900.0	1300.0
11	11.1	5.1	985.0	1215.0
12	11.0	5.0	1068.3	1131.7
13	10.9	4.9	1150.0	1050.0
14	10.8	4.8	1230.0	970.0
15	10.7	4.7	1308.3	891.7
16	10.6	4.6	1385.0	815.0
17	10.5	4.5	1460.0	740.0
18	10.4	4.4	1533.3	666.7
19	10.3	4.3	1605.0	595.0
20	10.2	4.2	1675.0	525.0
21	10.1	4.1	1743.3	456.7
22	10.0	4.0	1810.0	390.0

23	9.9	3.9	1875.0	325.0
24	9.8	3.8	1938.3	261.7
25	9.7	3.7	2000.0	200.0
26	9.6	3.6	2060.0	140.0
27	9.5	3.5	2118.3	81.7
28	9.4	3.4	2175.0	25.0
29	9.3	3.3	2230.0	-30.0 (cut-off, land)

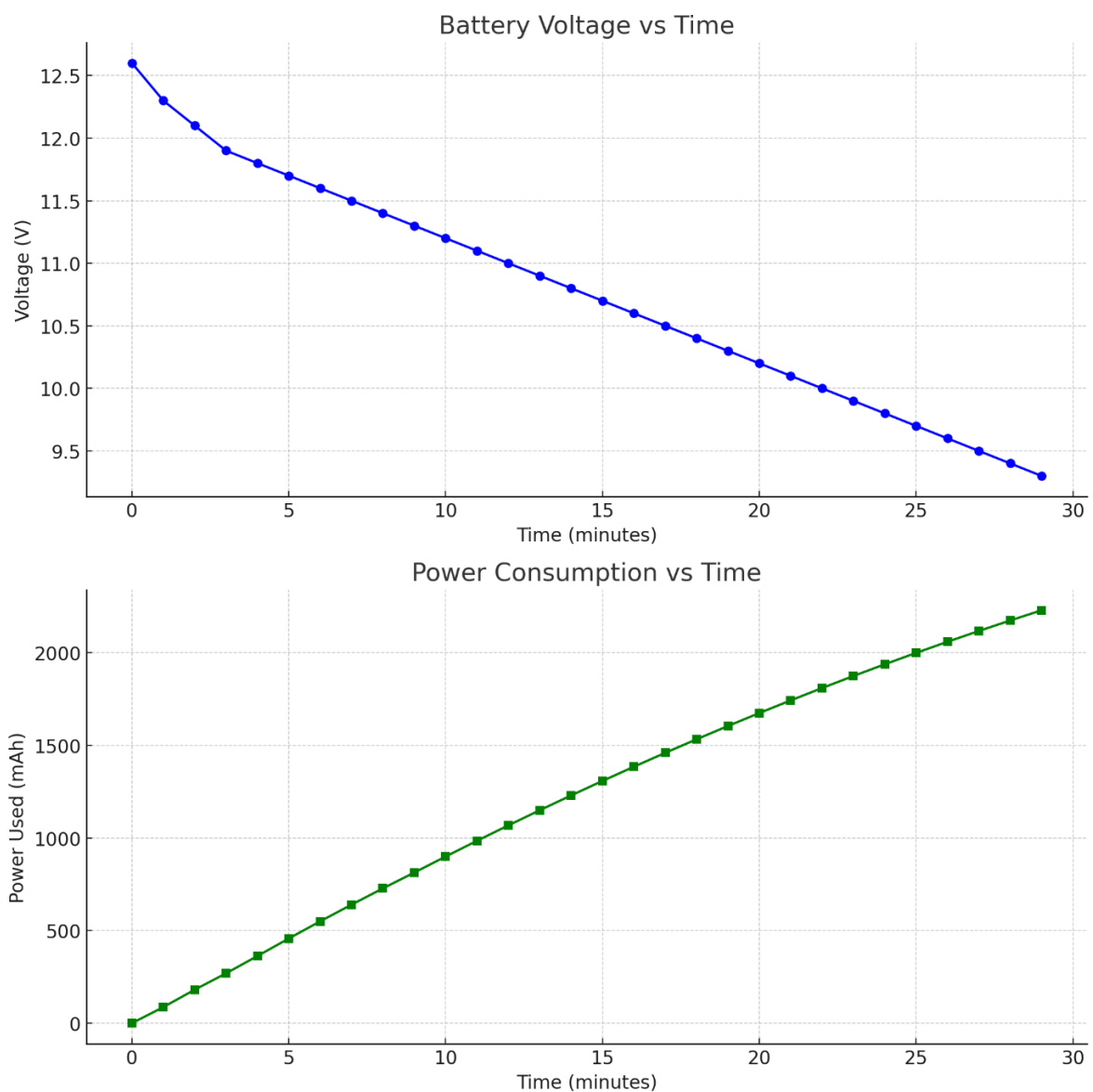


Fig 7.3.1 graph showing the battery voltage vs time and power consumption vs time

7.4 YOLO Detection Output Analysis

7.4.1 Fire detection :



Fig 7.4.1.1 (a) the image of fire before detection (b) the image of fire after detection

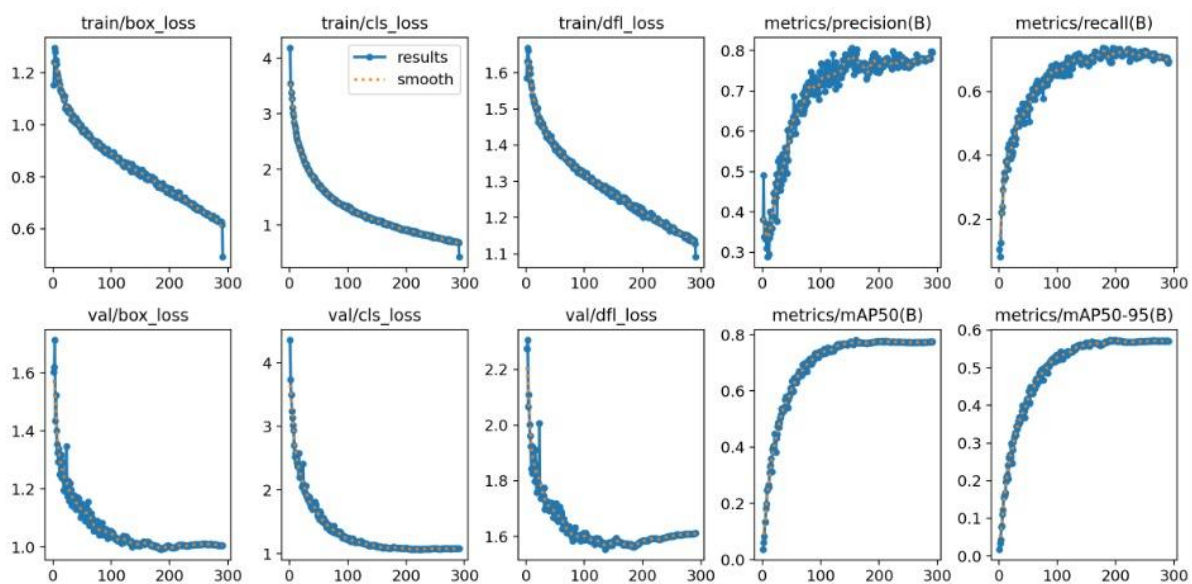


Fig 7.4.1.2 the training accuracy and loss curves of the trained fire model

7.4.2 Animal detection :

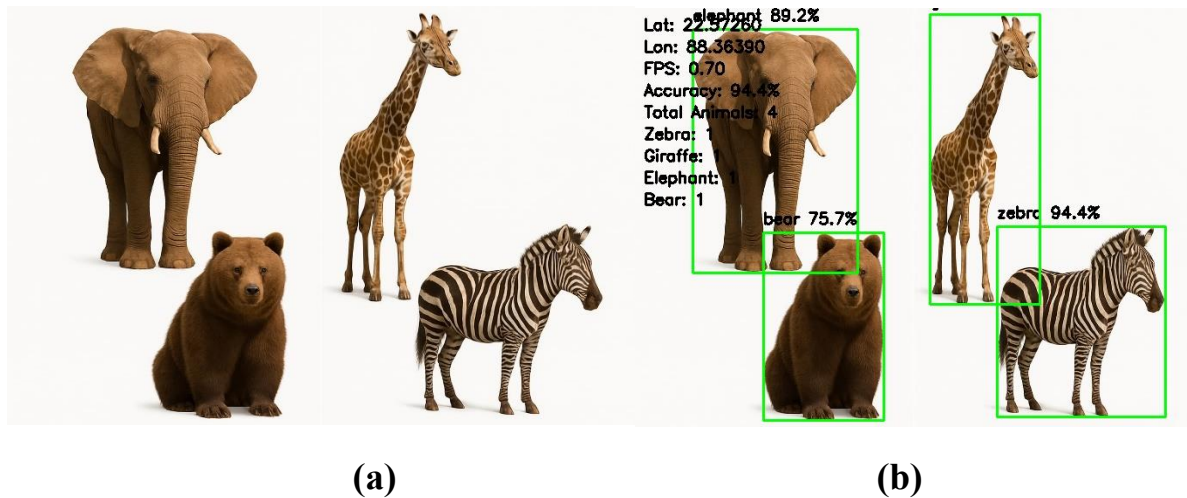


Fig 7.4.2.1 (a) the image of animal before detection (b) the animal images after detection

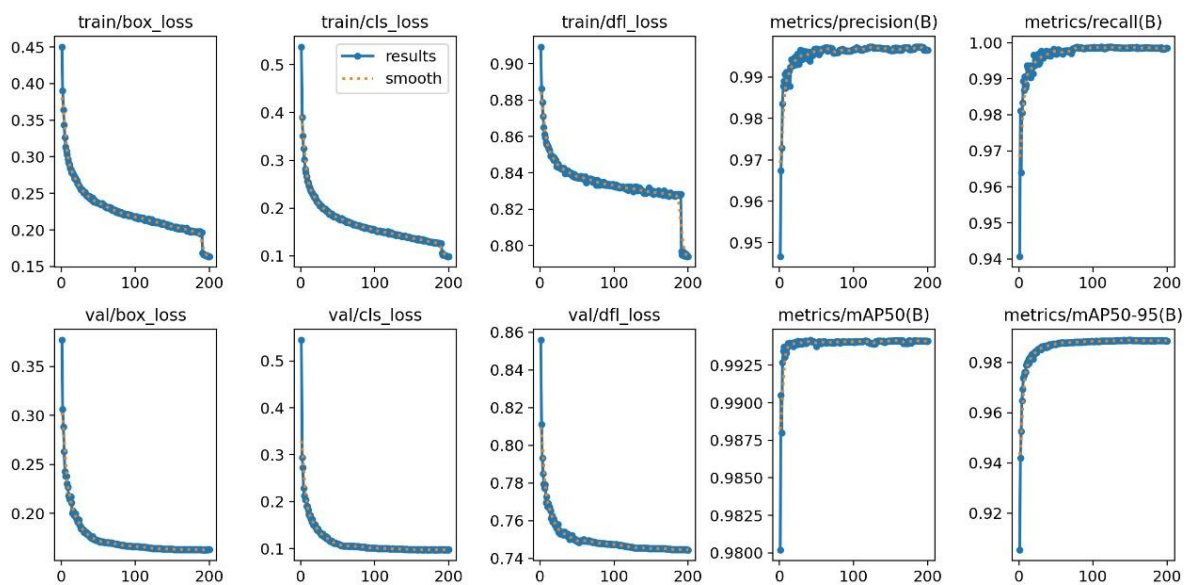
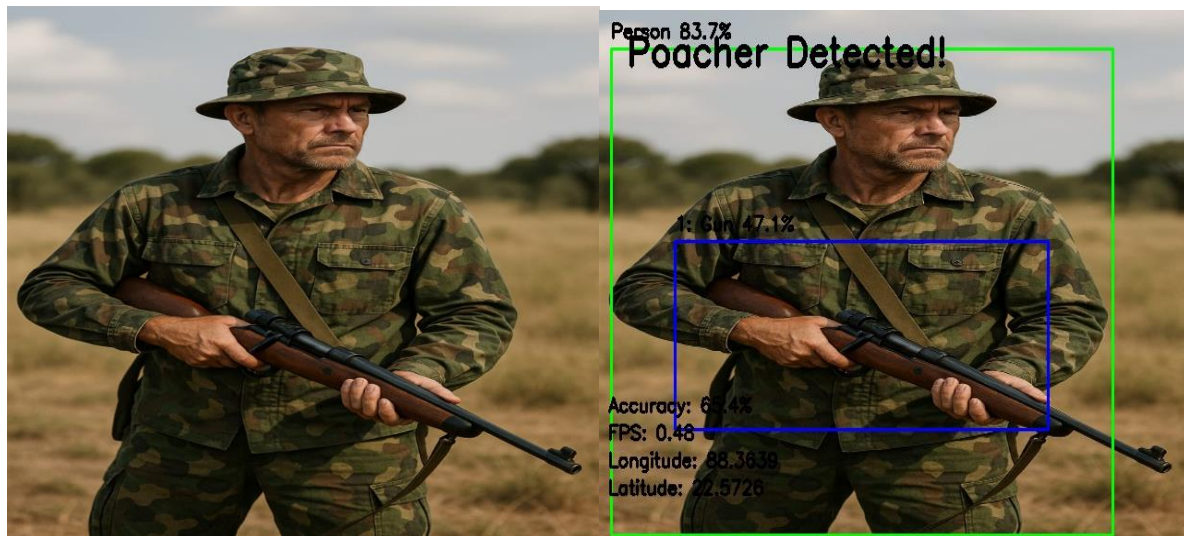


Fig 7.4.2.2 the training accuracy and loss curves of the trained animal model

7.4.3 Poacher detection :



(a)

(b)

Fig 7.4.3.1 (a) the image of poacher before detection (b) the image of poacher after detection

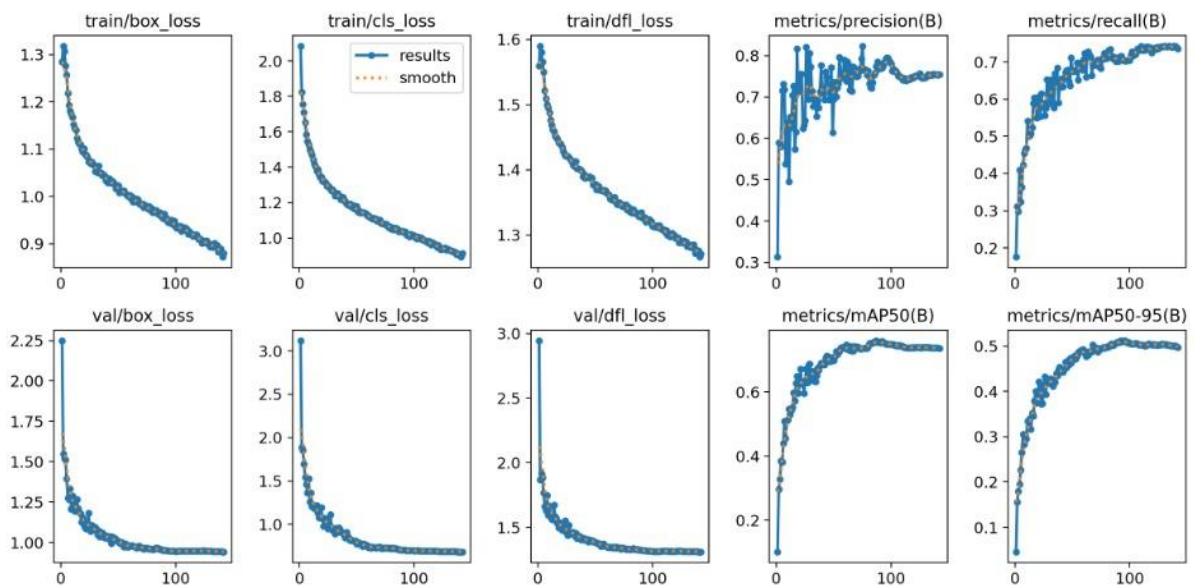


Fig 7.4.3.2 the training accuracy and loss curves of the trained poacher model

CHAPTER 8

APPLICATIONS AND USE CASES

8.1 Forest Fire Detection

Forest fire detection is an important application for drone-based computer vision systems. The drone's onboard camera takes live video feed, which is then analyzed using a YOLOv8 fire detection model. If fires or smoke are detected, the system immediately sends notifications, overlays bounding boxes on the affected area, and takes a photo with GPS coordinates. This allows forest officials to respond to fires more rapidly, especially in distant locations where ground patrol may be delayed. The technology reduces the need for continuous manual monitoring by automating the detection process. It also works in low-light or windy settings, depending on the camera's characteristics. Regular drone patrols utilizing this model aid in the early detection of fire threats, hence decreasing large-scale environmental damage, animal loss, and air pollution.

8.2 Illegal Logging Monitoring

Illegal logging harms forest ecosystems and biodiversity. Our drone system addresses this issue by monitoring and detecting illicit human activity in wooded areas using AI-based object detection technology. The drone, equipped with a YOLOv8 model trained to recognize individuals and weapons/tools (such as axes or chainsaws), hovers over defined regions and alerts authorities to suspicious behavior. When discovered, it notes the precise position with latitude and longitude, takes a visual snapshot, and sends an alert. This proof can help forest authorities take legal action and implement preventive measures. The drone's capacity to access difficult-to-reach areas provides a considerable advantage over manual patrols. Furthermore, it can patrol often and at regular intervals without tiring.

8.3 Wildlife Tracking

Wildlife tracking is critical for biodiversity conservation, and our method allows for non-invasive monitoring of animals in their natural habitat. The YOLOv8 animal identification model is used by the drone to distinguish and categorize animals such as deer, elephants, and other wildlife. When an animal is spotted, the system records the frame, tags it with the current GPS coordinates, and saves it with a timestamp. This enables academics and forest officials to monitor migration paths, habitat preferences, and population counts. Unlike traditional tracking methods, which require tagging or human presence, aerial monitoring causes minimum disruption to the animals. In the event of endangered animals, the system can send out notifications to conservationists. It also helps us understand how human encroachment and climate change affect animal behavior.

8.4 Disaster Area Surveillance

Wildlife tracking is critical for biodiversity conservation, and our method allows for non-invasive monitoring of animals in their natural habitat. The YOLOv8 animal identification model is used by the drone to distinguish and categorize animals such as deer, elephants, and other wildlife. When an animal is spotted, the system records the frame, tags it with the current GPS coordinates, and saves it with a timestamp. This enables academics and forest officials to monitor migration paths, habitat preferences, and population counts. Unlike traditional tracking methods, which require tagging or human presence, aerial monitoring causes minimum disruption to the animals. In the event of endangered animals, the system can send out notifications to conservationists. It also helps us understand how human encroachment and climate change affect animal behavior.

8.5 Smart Forest Management Systems

A smart forest management system automates and optimizes conservation activities by combining artificial intelligence, drone surveillance, GPS, and real-time data analytics. Our drone-based technology helps by constantly monitoring fires, poacher activity, illegal logging, and animal presence. All detections are captured using GPS-tagged images, resulting in a comprehensive database of forest occurrences. Authorities may use this unified data to identify patterns, assign resources, and plan targeted patrols. The method aids in the creation of heat maps of high-risk areas, tracking seasonal wildlife migration, and detecting patterns in unlawful activities. It also decreases human labor and mistake by eliminating the need for manual surveys. The solution speeds up on-site decision-making by merging edge computing and AI inference.

CHAPTER 09

CONCLUSION AND FUTURE WORK

9.1 Conclusion

An important milestone in the development of an intelligent, autonomous aerial system has been reached with the successful integration of the APM 2.8 flight controller with many crucial drone components. As the foundation of the drone's navigation and propulsion, the schematic designed for this drone guarantees effective communication between crucial modules such as the GPS (NEO M8N), ESCs with BLDC motors, FSi6 receiver, and power module. In addition, a separate ESP32 microcontroller acts as a backup unit for environmental sensing with the DHT11 and MQ8 sensors, adding more levels of data collection without affecting the main flying functions.

The drone's modularity and dependability are improved by this distributed system design. The ESP32 may function independently, recording data and supplying power to the ESP32-CAM for visual surveillance, by severing the connection between the flight controller and environmental sensing. In order to provide a steady voltage supply to every module, the power distribution system has been meticulously designed. In particular, delicate sensor circuits have been isolated from high-current motor routes to prevent noise and interference.

The drone can now fly steadily and independently thanks to its technology, and it can also intelligently monitor the environment and atmospheric conditions. Because of its scalability and adaptability, the system may incorporate future modules such as fire or poacher detection without requiring a redesign of its fundamental architecture. All things considered, this project successfully combines modular expandability, real-time data collecting, and precise control with embedded technologies in UAV design. It serves as a thorough paradigm for creating intelligent drones for study or surveillance in practical settings.

References

- [1] Air Shepherd. [Online]. Available: <https://airshepherd.org/>
- [2] Rainforest Connection, “Protecting forests with sound and AI,” [Online]. Available: <https://rfcx.org/>
- [3] Forest Survey of India, “Use of UAVs for Forest Monitoring,” Ministry of Environment, Forest and Climate Change, Govt. of India, 2021.
- [4] D. Floreano and R. J. Wood, “Science, technology and the future of small autonomous drones,” *Nature*, vol. 521, pp. 460–466, 2015.
- [5] SMART Partnership. “SMART (Spatial Monitoring and Reporting Tool).” [Online]. Available: <https://smartconservationtools.org/>
- [6] Air Shepherd. “Using Drones to Protect Elephants and Rhinos.” [Online]. Available: <https://airshepherd.org/>
- [7] DETECTOR Project, European Commission. “Drone-based Early Detection of Forest Fires.” [Online]. Available: <https://cordis.europa.eu/project/id/776528>
- [8] DroneSeed. “Drone-Based Reforestation.” [Online]. Available: <https://droneseed.com/>
- [9] Zhang, T., Li, J., & Huang, X. (2021). “Wildlife Detection in UAV Images Using YOLOv4,” *Remote Sensing*, vol. 13, no. 4, pp. 622–634. DOI: 10.3390/rs13040622
- [10] L. P. Koh and S. A. Wich, “Dawn of drone ecology: low-cost autonomous aerial vehicles for conservation,” *Tropical Conservation Science*, vol. 5, no. 2, pp. 121–132, 2012.
- [11] C. Zhang and J. M. Kovacs, “The application of small unmanned aerial systems for precision agriculture: a review,” *Precision Agriculture*, vol. 13, no. 6, pp. 693–712, 2012.
- [12] British Antarctic Survey. “Monitoring Glaciers and Wildlife with UAVs.” [Online]. Available: <https://www.bas.ac.uk/>
- [13] K. Muhammad, J. Ahmad, S. Mehmood, and S. W. Baik, “Convolutional neural networks-based fire detection in surveillance videos,” *IEEE Access*, vol. 6, pp. 18174–18183, 2018.
- [14] H. Liu et al., “A deep learning-based forest fire detection approach using UAV and YOLOv4,” *Remote Sensing*, vol. 12, no. 21, pp. 1–17, 2020. DOI: 10.3390/rs12213449.

- [15] D. Chavan and A. Shirsat, "Forest Fire Detection using Machine Learning Techniques," *Procedia Computer Science*, vol. 171, pp. 1339–1348, 2020.
- [16] H. Liu et al., "A deep learning-based forest fire detection approach using UAV and YOLOv4," *Remote Sensing*, vol. 12, no. 21, pp. 1–17, 2020.
- [17] L. P. Koh and S. A. Wich, "Dawn of drone ecology: low-cost autonomous aerial vehicles for conservation," *Tropical Conservation Science*, vol. 5, no. 2, pp. 121–132, 2012.
- [18] DroneSeed. "Drone-Based Reforestation." [Online]. Available: <https://droneseed.com/>